OCI Autonomous Database Serverless

Technical Architecture Connectivity and Security Patterns

ACE Program

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1. Service Overview

OCI Autonomous Database Serverless (ADB-S) is a fully managed PaaS service by Oracle Cloud Infrastructure that enables provisioning, managing, and operating highly secure, scalable, and autonomous Oracle databases, minimizing the operational activities typically required in traditional database management. The service provides advanced security, automation, elastic scalability, and high availability capabilities, designed to support enterprise application workloads in both cloud-native contexts and in integration with legacy and on-premise systems. ADB-S supports enterprise and cloud-native workloads, integrating natively with OCI security, networking, and identity services.

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/index.html

1.1 Autonomous Provisioning and Management

ADB-S automates the main database operational activities: (i) instance provisioning and configuration; (ii) automatic patching of the database and underlying infrastructure; (iii) automatic backup and recovery management; (iv) automatic performance tuning. These capabilities significantly reduce operational risk and human error.

1.2 Integrated Security by Design

The service implements a security-by-default model including: encryption of data at-rest and in-transit (TCPS/mTLS); native integration with OCI Identity and Access Management (IAM); support for authentication via IAM Database Token; database isolation via Private Endpoint within a VCN. Security is managed at multiple layers: network, identity, database, and auditing.

1.3 Elastic-Dynamic Scalability

ADB-S supports automatic and on-demand scaling of compute resources — CPU, memory, I/O. Capacity is dynamically adapted based on application workload, without manual intervention or downtime. With auto-scaling enabled, the database can automatically increase ECPUs beyond the configured base value up to the maximum limit. Additional resources are billed only for the period in which they are actually used, with sub-hourly granularity based on effective average consumption.

1.4 Integration with OCI Services

ADB-S integrates natively with: Compute Instances, Oracle Functions, OCI API Gateway, OCI Data Safe, OCI Logging and Monitoring. This integration enables the creation of complete, secure, and consistent application architectures within the Oracle Cloud ecosystem.

1.5 Logging, Auditing and Monitoring

The service provides: (i) database Unified Audit Trail for tracking SQL and security operations; (ii) OCI Audit Logs for control plane operation tracking; (iii) operational metrics available via OCI Monitoring. The service can also be integrated with OCI Data Safe for advanced activity monitoring, security risk analysis, and compliance support.

2. Service Components

The service components are substantially as follows:

2.1 Autonomous Database Serverless (ADB-S)

PaaS resource representing the fully managed Oracle database instance, responsible for application workload execution and autonomous management of: provisioning, patching, backup, performance tuning, and high availability.

2.2 OCI Control Plane

Exposes service management capabilities (create, scale, configure, lifecycle operations) via: OCI Console, OCI CLI, OCI SDK, OCI REST API. Access is governed by OCI Identity and Access Management (IAM) and is tracked via OCI Audit Logs.

2.3 Data Plane

The Data Plane manages access to the Autonomous Database application data. The service supports connections via TCPS and mTLS, both usable for database access. By default, both modes are enabled. Via configuration at the Autonomous Database level, exclusive mTLS use can be enforced by disabling standard TCPS connections, based on security and compliance requirements.

2.4 Network Integration — Private Endpoint

ADB-S can be configured with a Private Endpoint within a Virtual Cloud Network (VCN). This configuration enables: private database access; integration with Dynamic Routing Gateway (DRG); hybrid connectivity with on-premise environments; absence of public exposure of the database endpoint.

2.5 Logging, Auditing and Monitoring

The service integrates natively with OCI observability and security services: Unified Audit Trail for SQL and security operations tracking; OCI Audit Logs for Control Plane operation tracking; OCI Monitoring for operational and capacity metrics. OCI Data Safe provides advanced database activity monitoring, security risk analysis, and compliance support.

2.6 Storage and Backup

ADB-S manages storage and backup operations fully automatically. Storage is serverless and scalable. Backups are automatic and encrypted. Point-in-time recovery is supported. No manual configuration is required from the user. Backup and recovery capabilities are part of the service control plane management.

3. Use Cases and Reference Scenario

In the reference scenario, OCI Autonomous Database Serverless is used as the central data store of the architecture, accessible exclusively via private endpoint by applications in OCI and on-premise environments. The scenario highlights database access by different application workloads using private connectivity and encrypted traffic, with no public exposure of the database endpoint. The service enables enterprise database use in a secure and managed mode. Some typical use cases are:

3.1 Database for Enterprise Applications

ADB-S provides a managed Oracle database supporting business-critical applications, both cloud-native and legacy, ensuring security, scalability, and high availability.

3.2 Cloud and Hybrid Architecture Support

The database can be used by applications hosted on OCI, on-premise, or other cloud environments, through private and secure connectivity.

3.3 Integration with OCI Application Services

ADB-S can be integrated with Compute Instances, Oracle Functions, OKE, and API Gateway as the central data layer of the application architecture.

4. Network Architecture — Scenario Analysis

4.1 Introduction

This section describes in full detail the two reference architectural scenarios analyzed, tested in the field, and documented. Both scenarios use ADB-S with Private Endpoint in a private VCN subnet (<subnet-cidr>). The fundamental difference between the two scenarios lies in the Public Access configuration and the consequent network flow model. Sections 4.3 and 4.4 contain the complete textual description of all four diagrams present in the original presentation (slides 5, 6, 7, and 14). Each diagram is described exhaustively — every component, every arrow, every annotation, every flow both allowed and blocked — so that the full technical content of the visual can be reconstructed from text alone.

4.2 Conceptual Distinction — Private Endpoint vs Service Gateway

Before proceeding with scenario description, it is essential to establish the conceptual distinction between the two OCI network elements involved.

The Private Endpoint (PE) is an Oracle-managed VNIC with a private IP within the customer VCN subnet. It is the actual network interface of the ADB-S database within the VCN. All SQL*Net traffic (TCPS 1521 / mTLS 1522) is handled exclusively by this interface. The PE is the unique, non-bypassable access point for all SQL database traffic.

The Service Gateway (SGW) is a VCN-level gateway that allows resources within the VCN to reach Oracle services exposed on the Oracle Services Network (OSN) without transiting the public internet. The SGW handles traffic toward OSN endpoints including the ADB-S HTTPS services (APEX, ORDS, Database Actions). The SGW is structurally unable to carry SQL*Net traffic toward an ADB-S with Private Endpoint active: the SQL*Net listener on the PE rejects with ORA-12529 any connection attempt via the public OSN endpoint.

These two traffic planes — SQL*Net via PE and HTTPS via SGW/OSN — are strictly separated and operate under completely independent rules. Confusing the two is the most common source of architectural misconfigurations.

4.3 Diagrams Description — Slides 5, 6, 7 (Scenario A Network Flows)

4.3.1 Slide 5 — "OCI Network Flows: Service Gateway vs Private Endpoint (ADB-S)"

The diagram is titled "OCI Network Flows - Service Gateway vs Private Endpoint (ADB-S)" and illustrates the complete network topology of the ADB-S deployment with Private Endpoint enabled, showing simultaneously all possible connection flows — both allowed and blocked. A legend in the lower-right corner defines two markers: [OK] in green for connections that are allowed and functional, and [X] in red for connections that are rejected or blocked. The diagram is organized across three vertical levels: the VCN with its internal subnet at the top; the Oracle Services Network (OSN) to the right and below; and annotation boxes at the bottom.

VCN and Subnet Topology

The entire diagram is contained within an outer box labeled "VCN: <your-vcn-name>". Inside this VCN, a clearly delimited subnet box is labeled "Subnet privata <subnet-cidr>". Within this subnet, three compute entities are visible. The first is the Compute VM (<compute-vm-ip>, network interface enp0s5), shown in the upper-left of the subnet. The second is the Gateway logico VCN (the VCN default gateway, <vcn-gw-ip>), shown in the upper-center of the subnet. The third is the Service Gateway (sgw-private), shown at the upper-right, outside the subnet box but still within the VCN boundary. The routing flow from the Compute VM to the VCN Gateway is labeled "Default route → <vcn-gw-ip>". The flow from the VCN Gateway to the SGW is labeled "Private routing to SGW". This routing path exists for all traffic that exits the subnet toward Oracle Services Network via the SGW.

ADB-S Private Endpoint Block

The ADB-S Private Endpoint is shown as a distinct box in the lower-left of the subnet area, rendered with a light purple/lavender background to distinguish it visually from compute resources. The block displays three key data points: the internal private IP (<adb-pe-ip>), labeled as "<adb-pe-ip> internal IP" in the original; the SQL*Net hostname <adb-prefix>.adb.eu-<region>-1.oraclecloud.com; and the HTTPS hostname <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com. Below the PE block is an annotation note with three bullet points confirming the behavior of this endpoint: "Same private IP for *.oraclecloud.com and *.cloudapps.com" — meaning both hostnames resolve to the same PE private IP within the VCN DNS; "TCPS 1521 / mTLS 1522 → [OK] Works internally" — confirming that SQL*Net is fully functional when targeting the PE directly; "HTTPS 443 → [OK] Works internally" — confirming that HTTPS tools are also functional when targeting the PE directly.

Direct Flows from Compute VM to ADB-S Private Endpoint

Two separate arrows depart from the Compute VM and curve downward toward the ADB-S Private Endpoint. Both are drawn in blue (the color convention for allowed connections in this diagram) and both are labeled [OK] in green. The first arrow covers SQL*Net and is annotated "[OK] TCPS (1521 / 1522) SQL*Net / mTLS". The second arrow covers HTTPS tools and is annotated "[OK] HTTPS 443 APEX / ORDS". These two flows represent the primary operational access paths for applications residing within the same VCN: they go directly from the Compute VM to the PE private IP (<adb-pe-ip>) without involving any gateway, any SGW, or any OSN endpoint. These connections are fully functional and are the design-intended access paths for intra-VCN workloads.

SGW Flows Toward OSN — Key Behavioral Split

From the SGW, two arrows depart toward the Oracle Services Network block. This is the central behavioral split illustrated by the diagram. The first arrow is drawn in red and carries the label "[X] TCPS (1521 / mTLS 1522) ORA-12529 - Listener refused". This arrow is directed toward the "ADB Public SQLNet Endpoint" block within OSN. The red color and the [X] marker confirm that this flow is always rejected. The second arrow is drawn in green and carries the label "[OK] HTTPS 443 via Service Gateway". This arrow is directed toward the "ADB HTTPS Endpoint (APEX/ORDS)" block within OSN. The green color and the [OK] marker confirm that HTTPS toward the authorized long-form HTTPS domain is allowed via SGW when the traffic originates from the same VCN as the PE.

Oracle Services Network (OSN) Block

The OSN block is drawn as a large box in the lower-right of the diagram, labeled "Oracle Services Network (OSN) Region eu-<region>-1 CIDR <oracle-osn-ip>/16". Inside this block, two endpoint boxes are visible. The first, with a darker blue background, is labeled "ADB Public SQLNet Endpoint adb.eu-<region>-1.oraclecloud.com IP <oracle-osn-ip> (TCPS 1521 / mTLS 1522)". This is the target of the blocked red arrow from the SGW, confirming that this endpoint — the public SQL*Net endpoint on OSN — is structurally unreachable for SQL*Net traffic when the database has a Private Endpoint active. The second, with a lighter blue background, is labeled "ADB HTTPS Endpoint (APEX/ORDS) <adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com IP <oracle-osn-ip> / <oracle-osn-ip> (HTTPS 443)". This is the target of the allowed green arrow from the SGW, confirming that HTTPS access to ADB-S tools via the authorized long-form HTTPS domain is functional through SGW and OSN.

Bottom Annotation Notes

Three annotation boxes at the bottom of the diagram provide additional context for each component. The first annotation, below the ADB-S Private Endpoint block, is titled "ADB Private Endpoint" and summarizes: "Same private IP for *.oraclecloud.com and *.cloudapps.com"; "TCPS 1521 / mTLS 1522 → [OK] Works internally"; "HTTPS 443 → [OK] Works internally". The second annotation, below the ADB Public SQLNet Endpoint box in OSN, is titled "ADB SQLNet FQDN via SGW (OSN)" and summarizes: the hostname is adb.eu-<region>-1.oraclecloud.com; the IP is <oracle-osn-ip>; "TCPS 1521 / mTLS 1522 → [X] Rejected by listener"; "Confirmed after ACL refresh and Allow Public Access re-enable" — this last point is critical: the block is permanent and persists even after re-enabling Allow Public Access or refreshing ACLs. The third annotation, below the ADB HTTPS Endpoint box in OSN, is titled "ADB HTTPS FQDN via SGW (OSN)" and summarizes: the hostname is <adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com; the IPs are <oracle-osn-ip> and <oracle-osn-ip>; "HTTPS 443 → [OK] Supported (APEX / ORDS)"; "Routed via OSN through SGW (private)".

Summary of Behavioral Findings from Slide 5 Diagram

- [OK] SQL*Net TCPS 1521 / mTLS 1522 from Compute VM directly to PE <adb-pe-ip>: allowed
- [OK] HTTPS 443 from Compute VM directly to PE <adb-pe-ip>: allowed
- [X] SQL*Net TCPS 1521 / mTLS 1522 via SGW → OSN endpoint adb.eu-<region>-1.oraclecloud.com: blocked with ORA-12529 — structural and permanent, independent of ACL or public access settings
- [OK] HTTPS 443 via SGW → OSN endpoint <adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com: allowed from same PE VCN

4.3.2 Slide 6 — "Autonomous Database Serverless — Real Flows (Private + LB + OSN)"

The diagram is titled "Autonomous Database Serverless — Flussi Reali (Privato + LB + OSN)" and shows the complete operational flow architecture of Scenario A with all three access paths active simultaneously: direct PE access, LB L7-mediated SGW/OSN access, and the blocked "short URL" path. A legend at the bottom defines three color codes: blue (#blue) for direct private IP access without SGW; dark green (#darkgreen) for flow via SGW using the new authorized domain, resulting in HTTP 302→200 OK; red (#red) for flow via SGW using the old short domain, resulting in HTTP 404 blocked.

On-Premise / VPN Client Block

At the very top of the diagram is a green-bordered box labeled with the UML stereotype "OnPrem" and the title "On-Prem / VPN Client — PC aziendale (file hosts con IP privati)". Inside this box, a "Browser / JDBC Client" entity is shown. From this entity, three arrows depart, each toward a different access path, each drawn in a

different color corresponding to the legend. The first arrow, dark green, goes toward the LB L7 and is annotated "HTTPS 443 (<adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com)" — this is the long authorized HTTPS domain routed via LB and SGW. The second arrow, red, also goes toward the LB L7 but is annotated "HTTPS 443 (<adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com)" — this is the short "private" URL that will be blocked by the Oracle proxy after ~1 hour. The third arrow, blue with dashed line style, bypasses both LB and SGW entirely and is annotated "HTTPS / TCPS (File hosts → IP privato <adb-pe-ip>) Tools e JDBC funzionano (No SGW, no DRG)" — this is the direct path using the hosts file on the on-premise client to resolve both ADB-S hostnames to the PE private IP.

ACL Annotation Box

A yellow annotation box positioned to the right of the VCN boundary reads: "ACL su 'Enable Public Access' → Solo subnet <subnet-cidr> ammessa → Nessun accesso da Internet". This confirms that in Scenario A, the "Allow Public Access" ACL is configured to admit exclusively the VCN private subnet, effectively preventing any internet-originated access to the public Oracle endpoint even though public access is technically enabled. This is the defining security control of Scenario A.

VCN and Internal Components

The main VCN box (<your-vcn-name>, subnet <subnet-cidr>) contains three primary entities. The first is the Load Balancer L7 block, shown in a blue box, labeled "Load Balancer L7 (HTTPS 443) Frontend <lb-frontend-ip> Backend → OSN <oracle-osn-ip>". The LB L7 terminates TLS on the frontend, re-originates the connection toward OSN with the VCN local IP, and logs the real client IP. The second entity is the ADB Private Endpoint block, showing IP <adb-pe-ip>, SQL*Net hostname <adb-prefix>.adb.eu-<region>-1.oraclecloud.com (→ TCPS 1522), and HTTPS hostname <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com (HTTPS OK). The third entity is the Service Gateway (SGW → Oracle Services Network), positioned below the LB L7. Two arrows connect the LB L7 downward to the SGW: one dark green labeled "Routed via SGW", one red also labeled "Routed via SGW". Both the authorized and the blocked HTTPS flows transit through the LB L7 and then through the SGW toward OSN.

Oracle Services Network — Oracle Public Proxy — Core Discrimination Mechanism

The OSN block at the bottom of the diagram, labeled "Oracle Services Network (OSN)", contains a single central entity: "<oracle-osn-ip> - Oracle Public Proxy (Filtra domini ADB autorizzati)" — the Oracle Public Proxy at IP <oracle-osn-ip> which filters authorized ADB domains. This is the key architectural element that determines which HTTPS requests are allowed and which are blocked. From this proxy, two arrows depart: a dark green arrow going toward "Nuovo dominio OK: <adb-https-prefix>.adb..." labeled "HTTP 302 → 200 OK (Accesso Tools/ORDS consentito)"; and a red arrow going toward "Vecchio dominio bloccato: <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com" labeled "HTTP 404 - dominio non autorizzato". The proxy is connected to OSN via two labeled flows: "Proxy autorizzato" (the dark green connection from SGW to proxy) and "Proxy OSN riceve richiesta" (for the red blocked connection).

This is the fundamental mechanism: both the long domain (<adb-https-prefix>...) and the short domain (<adb-prefix>...) resolve to the same PE private IP (<adb-pe-ip>) via VCN internal DNS. However, when a request transits via LB L7 → SGW → OSN and arrives at the Oracle Public Proxy, the proxy performs domain-based filtering. The long HTTPS domain (<adb-https-prefix>...) is in Oracle's list of authorized domains for that database on OSN, so the proxy responds HTTP 302→200 OK and the request succeeds. The short "private" domain (<adb-prefix>...) is NOT included in the authorized domain set on the proxy for that database, so after approximately 1 hour from provisioning (when Oracle's internal configuration fully propagates), the proxy returns HTTP 404. The label "private" in the Oracle console refers to the internal routing, not to access restriction, and this domain was never intended for use via the OSN/SGW path beyond the initial ~1-hour provisioning window.

Blue Direct Path — File Hosts Bypass

The blue dashed arrow from the on-premise client goes directly to the ADB Private Endpoint block, bypassing the LB L7, the SGW, and the OSN entirely. This path functions because the on-premise client's hosts file is configured to resolve both ADB-S hostnames — SQL*Net and HTTPS — to the PE private IP (<adb-pe-ip>). Via VPN+DRG, the client can reach this IP directly, without any Oracle Services Network involvement. Neither the SGW nor the Oracle Public Proxy are in this path, so the proxy domain-filtering constraint does not apply. Both the browser (for HTTPS APEX/ORDS access) and the JDBC client (for SQL*Net mTLS access) function correctly on this path.

Summary of Behavioral Findings from Slide 6 Diagram

- [OK] HTTPS via LB L7 → SGW → OSN using long domain <adb-https-prefix>...: allowed, Oracle Public Proxy returns HTTP 302→200 OK
- [BLOCKED after ~1h] HTTPS via LB L7 → SGW → OSN using short domain <adb-prefix>...: Oracle Public Proxy returns HTTP 404 — domain not in authorized set
- [OK] HTTPS/TCPS directly to PE <adb-pe-ip> via file hosts + VPN: allowed — no proxy, no SGW, no OSN involved

- ACL on Allow Public Access: only <subnet-cidr> admitted — no internet access possible

4.3.3 Slide 7 — "Scenario A: ADB-S PE + Allow Public Access ON + ACL — VCN: <your-vcn-name>"

The diagram is titled "SCENARIO A — ADB-S: PE + Allow Public Access ON + ACL — VCN: <your-vcn-name>" and represents the complete Scenario A architecture with all its components, flows, and cross-VCN enforcement. A color and style legend in the lower-right corner defines: solid dark green arrow for connections allowed and functional; red dashed arrow for connections blocked or rejected; red/orange fill for ADB-S Private Endpoint; blue fill for Load Balancer Layer 7; medium blue fill for Service Gateway; pink fill for OSN — Oracle Services Network; light blue fill for VCN diversa from the PE VCN. The outermost boundary of the diagram represents the full OCI IaaS region.

Two VCNs — PE VCN and VCN Diversa

Within the OCI IaaS region boundary, two VCN containers are visible side by side. On the left is the main PE VCN container, labeled "VCN: <your-vcn-name> — Subnet privata: <subnet-cidr>". On the right is a separate lighter-blue container labeled "VCN DIVERSA (altra VCN / altro tenant)" containing a single entity "Workload altra VCN + proprio SGW". From this "Workload altra VCN" entity, two red dashed arrows depart, both going left-to-right toward the OSN area and the PE VCN. The first arrow, labeled with a note icon, reads "HTTPS → HTTP 403 immediato — SQL*Net → ORA-12529 — 'same private network' enforcement — non bypassabile via SGW". Both blocked flows (HTTPS HTTP 403 and SQL*Net ORA-12529) are shown as a single double-annotation on one arrow set, confirming that any workload in a VCN other than the PE VCN that attempts to reach ADB-S — whether via its own SGW for HTTPS or via OSN for SQL*Net — receives an immediate block, regardless of whether its IP was added to the Allow Public Access ACL.

Main PE VCN — All Internal Components

Inside the main PE VCN, four main blocks are arranged. In the upper area: the Compute VM (<compute-vm-ip>, enp0s5), shown as a white box with a server icon. To the left-center: the Load Balancer L7 block (blue fill), labeled "Load Balancer L7 — HTTPS 443 — Frontend: <lb-frontend-ip> — Backend → OSN <oracle-osn-ip>". In the center: the ADB-S Private Endpoint block (red/orange fill), shown as the largest box, labeled with all four key parameters: "IP: <adb-pe-ip>; <adb-prefix>.adb.eu-<region>-1.oraclecloud.com; <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com; Allow Public Access = ON; ACL: solo <subnet-cidr>". In the lower-center: the Gateway logico VCN (<vcn-gw-ip>) connecting the VCN to the SGW. At the bottom: the SGW block (medium blue fill), labeled "SGW sgw-private → Oracle Services Network".

Internal Allowed Flows — Green Solid Arrows

From the Compute VM, a solid green arrow labeled "HTTPS 443" goes toward the LB L7 frontend, indicating that HTTPS traffic from the VM to the LB L7 is allowed. From the ADB-S PE block, a solid green arrow labeled "HTTPS 443 — <adb-prefix>...oraclecloudapps.com → <adb-pe-ip>" indicates that the short HTTPS URL resolved to the PE private IP is accessible directly from within the VCN. From the LB L7, a solid green arrow goes through the VCN Gateway toward the SGW, and then from the SGW a solid green arrow goes down toward the OSN, labeled "HTTPS 443 via SGW — <adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com — HTTP 302→200 OK". This represents the fully functional HTTPS path via LB L7 → VCN Gateway → SGW → OSN for the authorized long domain.

Internal Blocked Flows — Red Dashed Arrows

From the ADB-S PE block, a red dashed arrow labeled "TCPS 1521/mTLS 1522 via SGW" goes toward the SGW. Then from the SGW a red dashed arrow descends toward the OSN, arriving at the "ADB Public SQLNet" block and labeled "ORA-12529 — Listener rifiuta SQL da OSN — Strutturale — sempre". This confirms that SQL*Net via SGW/OSN is structurally rejected at all times regardless of configuration. Additionally, a red dashed arrow from the ADB-S PE block is labeled "<adb-prefix>...oraclecloudapps.com — HTTP 404 dopo ~1h — inibito da Oracle proxy/ACL", confirming the short URL behavior via SGW.

OSN — Three Distinct Destination Entities

Below the VCN is the OSN block (CIDR <oracle-osn-ip>/16) containing three separate entities. Entity 1 (dark red background): "ADB Public SQLNet — adb.eu-<region>-1.oraclecloud.com — IP: <oracle-osn-ip> — (TCPS 1521 / mTLS 1522)". This receives the always-blocked SQL*Net red arrow from the SGW. Entity 2 (orange background): "ADB HTTPS Endpoint (APEX/ORDS) — <adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com — IP: <oracle-osn-ip> / <oracle-osn-ip> — (HTTPS 443)". This receives the allowed green HTTPS arrow from the SGW. Entity 3 (orange-red background): "Oracle Public Proxy <oracle-osn-ip> — Filtra domini ADB autorizzati — <adb-prefix>... inibito dopo ~1h". This receives the red dashed arrow representing the short-URL 404 behavior.

Summary of Behavioral Findings from Slide 7 Diagram

- [OK] SQL*Net TCPS 1521 / mTLS 1522 from VCN Compute VM directly to PE <adb-pe-ip>: allowed
- [X] SQL*Net TCPS 1521 / mTLS 1522 via SGW → OSN: BLOCKED with ORA-12529 — structural, permanent
- [OK] HTTPS 443 via LB L7 → VCN GW → SGW → OSN long domain <adb-https-prefix>...: allowed — HTTP 302→200 OK
- [OK for ~1h, then X] HTTPS 443 via SGW short domain <adb-prefix>...: functional for ~1 hour, then Oracle Public Proxy returns HTTP 404
- [X immediate] HTTPS from VCN DIVERSA via own SGW: HTTP 403 — "same private network" enforcement — non-bypassable
- [X immediate] SQL*Net from VCN DIVERSA via own SGW: ORA-12529 — same enforcement
- Allow Public Access = ON + ACL restricted to subnet CIDR: effectively prevents any internet access while enabling internal VCN HTTPS flows

5. Scenario A — PE + Allow Public Access ON + SGW + LB L7 + Network Firewall

5.1 Configuration Summary

ADB-S provisioned with Private Endpoint in private VCN subnet (<subnet-cidr>), Allow Public Access enabled, ACLs restricting access exclusively from the VCN subnet. In this configuration the Public Endpoint does not represent real public exposure but a logically limited endpoint, not reachable from the Internet or unauthorized VCNs. Scenario fully analyzed, tested in the field, and documented. Enterprise reference model for environments requiring access to ADB-S HTTPS tools with full traceability, control, and traffic segregation.

5.2 SQL*Net Access — TCPS 1521 / mTLS 1522

Allowed exclusively via PE private IP (<adb-pe-ip>) from the same VCN. The Service Gateway is structurally and intentionally blocked for SQL*Net traffic: any connection attempt toward the public OSN endpoint adb.eu-<region>-1.oraclecloud.com via SGW is rejected by the PE listener with ORA-12529. This behavior is independent of ACL configuration or public access setting and is an integral part of the Oracle security model for ADB-S with Private Endpoint active.

5.3 HTTPS 443 Access — ADB Tools via LB L7 and Service Gateway

The public HTTPS endpoint (<adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com) on OSN is reachable via SGW exclusively from the same PE VCN, with public access ON and ACLs permitting. HTTPS traffic transits through the Layer 7 Load Balancer (frontend <lb-frontend-ip>, backend OSN <oracle-osn-ip>) which terminates TLS, re-originates the connection toward OSN with the local VCN IP, and logs the real client IP. This is a necessary requirement because the SGW structurally verifies that traffic toward OSN originates from the same VCN in which it is configured.

The URL <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com, labeled "private" in the Oracle ADB dashboard but with public IP on Oracle external DNS, is reachable directly to the PE private IP from within the VCN. When routed via SGW toward OSN, it is functional for approximately 1 hour after provisioning, after which the Oracle internal proxy on OSN blocks it with HTTP 404 because it is not in the set of authorized domains for that database on OSN. From different VCNs, blocking is immediate with HTTP 403: "This database uses a private IP address that is only accessible from a client that is in the same private network as the Autonomous Database."

5.4 Why the OCI Network Load Balancer Cannot Be Used

The OCI NLB operates at Layer 3/4 in non-proxy pass-through mode: it does not terminate TLS, does not decrypt, and cannot respond to the HTTPS TLS handshake. To force HTTPS traffic toward OSN via SGW, an Application Load Balancer Layer 7 (OCI Flexible LB) is required, which terminates TLS on the frontend, decrypts the request, re-originates toward OSN with the local VCN IP, and logs the real client IP.

5.5 Structural ORDS Limitation — Proxy IP in the Unified Audit Trail

All ADB-S tools — APEX, ORDS, Database Actions, SQL Developer Web — are delivered via ORDS, which operates as an HTTP proxy between the client and the Oracle Database kernel. This is a structural behavior of ORDS completely independent of the network path: whether the client reaches ORDS via LB L7/SGW/OSN or directly via PE private IP on port 443, the ADB-S Unified Audit Trail always records the IP of the ORDS process as source IP — never the real HTTP client IP. The Unified Audit Trail alone is therefore insufficient to guarantee network identity traceability for HTTPS sessions toward ADB-S tools in any architectural scenario.

> **WARNING** *The Unified Audit Trail alone is insufficient to guarantee network identity traceability for HTTPS sessions toward ADB-S tools in any architectural scenario. This is a fundamental ORDS architectural constraint, not a configuration issue.*

5.6 Best Practice — OCI Network Firewall with TLS Inspection

The correct architectural solution to guarantee full client IP visibility is to position an OCI Network Firewall with TLS inspection enabled in front of the SGW. The Network Firewall — transparent to the programmer and to the application — intercepts HTTPS traffic before it reaches ORDS, logs the real source IP of the client, inspects Layer 7 content, and applies granular control policies, structurally compensating the Unified Audit Trail limitation for HTTPS sessions. Positioning in front of the SGW is preferable to positioning in front of the PE because HTTPS traffic toward the tools is a specific and limited perimeter — exclusively ORDS — while the PE also manages SQL*Net mTLS traffic concurrently.

5.7 Forcing HTTPS Only via LB L7 — ACL Control

To ensure no client bypasses the LB L7 by accessing the PE directly on port 443, the "Allow Public Access" ACLs are configured with only the LB L7 frontend IP (<lb-frontend-ip>) as the allowed source. HTTPS directly from any VM to PE <adb-pe-ip>:443 is blocked by ACL because the source is not the LB; HTTPS via LB L7 passes because the LB re-originates with its frontend IP. The public access ACLs govern exclusively the Oracle public endpoint on OSN and do not affect the PE private IP or SQL*Net traffic. Access control to the private PE remains delegated to Security List and NSG on the VCN subnet.

5.8 ZPR — Zero Trust Packet Routing

OCI Zero Trust Packet Routing (ZPR) protects sensitive data from unauthorized access via intent-based security policies written for resources assigned security attributes. ZPR applies to ADB-S exclusively in Private Endpoint configuration. ZPR operates as an additional and complementary layer to NSG and Security List: for a packet to reach the target it must pass all NSG, Security List, and ZPR rules. If any rule does not allow the traffic, the packet is discarded. ZPR is offered at no additional cost for OCI. Even if network security rules appear to allow access, ZPR can still prevent the connection if no policy explicitly authorizes it — enormously reducing risks from forgotten firewall openings or partially reconfigured VMs.

```
# ZPR policy — SQL*Net from authorized workload in same VCN:
in networks:<your-vcn-name> VCN
allow app:<authorized-workload> endpoints
to connect to db:<adb-s-name> endpoints
with protocol='tcp/1522'

# ZPR policy — SQL*Net from on-premise via DRG (IP range):
in networks:<your-vcn-name> VCN
allow '<subnet-cidr>'
to connect to db:<adb-s-name> endpoints
with protocol='tcp/1522'
```

5.9 Cross-VCN Access — Constraint and Supported Patterns

As documented in the field tests (slides 8-12 of the original presentation), the "same private network" constraint applied by Oracle to the Private Endpoint is strict on the PE VCN: any attempt to access ADB-S HTTPS tools from a different VCN — even with its own Service Gateway and even if the source was explicitly included in the ACL allowlist — is blocked immediately with HTTP 403. Enforcement is not based solely on the IP allowlist but on the VCN network context of the Private Endpoint. Similarly, SQL*Net traffic from different VCNs cannot reach the PE via their own SGW toward OSN, being blocked with ORA-12529. Ne consegue che nessuna VCN diversa può usare il proprio SGW per raggiungere ADB-S — né per SQL*Net né per HTTPS.

For workloads in VCNs different from the PE VCN there are only two supported architectural patterns. Pattern 1 is direct connectivity to PE via network peering toward the PE VCN: via Local Peering Gateway (LPG) for VCNs in the same region and tenancy, or via Dynamic Routing Gateway (DRG) for hub-spoke connectivity or for on-premise via FastConnect/IPSec VPN. Traffic reaches the PE private IP directly. Pattern 2 is transit via an intermediate hop in the PE VCN that changes the source IP: a workload in a different VCN routes toward the PE VCN via LPG or DRG, where an intermediate component — typically a LB L7 for HTTPS or a proxy/jump host for SQL*Net — receives traffic and re-originates it toward the PE or SGW with source IP belonging to the PE VCN subnet range. This IP source change is the structural requirement Oracle verifies. For SQL*Net, the source IP change via a hop in the PE VCN is not necessary since SQL*Net reaches the PE directly via LPG/DRG peering without passing through OSN.

> **NOTE** *No VCN other than the PE VCN can use its own SGW to reach ADB-S — neither for SQL*Net nor for HTTPS. The only SGW useful for HTTPS is that of the PE VCN, reachable from external VCNs only via an intermediate hop in the PE VCN that re-originates traffic with local IP. This is the fundamental architectural separation that distinguishes the role of SGW in the ADB-S network model with Private Endpoint.*

5.10 Conclusion — Scenario A

Scenario A is architecturally correct and valid as an enterprise reference for environments requiring access to ADB-S HTTPS tools with full traceability. Not adopted in the current deployment phase because ADB-S HTTPS tools (APEX, ORDS, Database Actions, SQL Developer Web) are not used in the operational architecture.

6. Scenario B — PE ONLY, No Public Access (Adopted Enterprise Scenario)

6.1 Diagram Description — Slide 14 — "Scenario B: ADB-S PE ONLY, Allow Public Access = OFF"

The diagram is titled "SCENARIO B — ADB-S: PE ONLY, Allow Public Access = OFF — Scenario Enterprise Finale — VCN: <your-vcn-name>" and represents the adopted final scenario. A color and style legend in the lower-right defines: solid green arrow for connections allowed and functional; red dashed arrow for connections blocked or non-applicable; red/orange fill for ADB-S Private Endpoint ONLY; blue fill for DRG — Dynamic Routing Gateway; green fill for SGW — present but not involved for ADB-S; light green fill for Private DNS Zone VCN Resolver; yellow/beige fill for On-Premise/VPN Client; pink fill for OSN — disabled for this database; light blue fill for VCN DIVERSA — non-applicable. The outermost boundary represents the OCI IaaS region.

OCI IaaS Region — Two VCNs and External Components

Within the OCI IaaS region boundary two VCN containers are visible. The main PE VCN (<your-vcn-name>, subnet <subnet-cidr>) is on the left. On the upper-right is the "VCN DIVERSA (altra VCN / altro tenant)" container (light blue fill), which contains a "Workload altra VCN + proprio SGW" entity. From this entity two red dashed arrows depart: one toward the OSN area annotated "SGW non coinvolto — nessun public endpoint attivo su OSN per questo DB"; one further right annotated "Non applicabile — nessun public endpoint attivo su OSN". Both confirm that VCN DIVERSA has no viable path toward this database, because no public endpoints were ever activated on OSN for this ADB-S instance. On the far right is the "ON-PREMISE / VPN Client — PC aziendale" block (yellow/beige fill) containing a "Browser / JDBC Client" entity, with the annotation "file hosts → IP privato <adb-pe-ip>".

Private DNS Zone — VCN Resolver Block

At the upper-left of the main PE VCN, a light green box labeled "Private DNS Zone — VCN Resolver" displays two DNS resolution entries: "<adb-prefix>.adb.eu-<region>-1.oraclecloud.com → <adb-pe-ip>" and "<adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com → <adb-pe-ip>". A note below states "(auto-provisioned da Oracle al deploy PE)". A gray dashed arrow from the DNS block points downward toward the ADB-S PE ONLY block, labeled "risolve → <adb-pe-ip>". This confirms that Oracle automatically provisions private DNS zones in the VCN Resolver at PE deployment time: both the SQL*Net hostname (*.oraclecloud.com) and the HTTPS hostname (*.oraclecloudapps.com) are resolved to the same PE private IP within the VCN, without any manual DNS configuration required by the customer.

SGW — Present in VCN but Not Involved for ADB-S

Within the main VCN a Service Gateway block (green fill) is visible, labeled "SGW (presente in VCN — NON coinvolto per ADB-S)". No connection arrow goes from the SGW toward the OSN or toward ADB-S in this scenario. The SGW is present in the VCN because it may serve other OSN services (e.g. Object Storage, OCI API Gateway), but for this specific ADB-S instance it has zero role. No active public endpoint exists on OSN for this database, so the SGW has no usable path toward ADB-S even if the VCN routing were configured to try.

Compute VM — Direct SQL*Net to PE

A white box labeled "Compute VM <compute-vm-ip>" is present inside the VCN. A solid green arrow labeled "TCPS 1521 / mTLS 1522 — diretto a <adb-pe-ip>" connects the Compute VM downward to the ADB-S Private Endpoint ONLY block. This represents the primary intra-VCN access path: direct SQL*Net from the Compute VM to the PE private IP. No LB L7, no SGW, no gateway is involved. The connection is direct within the same VCN, controlled exclusively by Security List, NSG, and ZPR policies on the PE subnet.

ADB-S Private Endpoint ONLY — Central Block

The central block of the diagram is the ADB-S Private Endpoint ONLY (red/orange fill), which is the largest and most visually prominent element. It shows: "IP: <adb-pe-ip>"; SQL*Net hostname "<adb-prefix>.adb.eu-<region>-1.oraclecloud.com"; HTTPS hostname "<adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com"; "Allow Public Access = OFF"; "Tool HTTPS non utilizzati". The "Allow Public Access = OFF" is the defining element of Scenario B: no public endpoint was ever initialized on OSN. There is no URL active on OSN for this database on any protocol. The PE is the unique and exclusive access point for all traffic — SQL*Net and HTTPS alike, if HTTPS were to be used.

OSN — Disabled for this Database

The OSN block (pink fill, CIDR <oracle-osn-ip>/16) appears to the right of center in the diagram. Inside the OSN block, a red box shows the public ADB-S endpoints: "adb.eu-<region>-1.oraclecloud.com — IP: <oracle-osn-ip>" and "<adb-https-prefix>.adb.eu-<region>-1.oraclecloudapps.com — IP: <oracle-osn-ip>". Below these entries a bold annotation reads: "MAI ABILITATI per questo ADB-S — Nessun listener attivo — Zero esposizione internet".

The OSN block itself receives only red dashed arrows — no green solid arrow arrives at OSN from the main VCN for this database. The OSN block exists in the diagram to show that these Oracle infrastructure endpoints exist at the platform level, but for this specific ADB-S instance they were never activated and no traffic is expected or possible toward them.

On-Premise / VPN Client — FastConnect / IPSec VPN + DRG

On the far right of the diagram, the "On-Premise / VPN Client — PC aziendale" block (yellow/beige fill) contains a "Browser / JDBC Client" entity with annotation "file hosts → IP privato <adb-pe-ip>". A solid green arrow goes from this block toward the "FastConnect / IPSec VPN" block (orange/yellow fill). Another solid green arrow connects FastConnect/VPN to the "DRG (Dynamic Routing Gateway)" block (blue fill) positioned within the main VCN boundary. From the DRG, a solid green arrow labeled "TCPS 1521 / mTLS 1522 — tramite FastConnect/VPN + DRG" connects downward to the ADB-S Private Endpoint ONLY block. This path represents the on-premise access flow: the corporate client connects via VPN or FastConnect, traffic arrives at the OCI DRG which routes toward the VCN subnet, and from there directly to the PE private IP. The client hosts file is configured to resolve both ADB-S hostnames to <adb-pe-ip>, bypassing the public Oracle DNS which would otherwise resolve them to OSN public IPs. Both the JDBC client (for SQL*Net mTLS) and the browser (for HTTPS if needed) are functional on this path.

Summary of All Flows from Slide 14 Diagram

- [OK] SQL*Net TCPS 1521 / mTLS 1522 from VCN Compute VM directly to PE <adb-pe-ip>: allowed
- [OK] SQL*Net TCPS 1521 / mTLS 1522 from on-premise via FastConnect/VPN + DRG to PE <adb-pe-ip>: allowed
- SGW: present in VCN for other OSN services — has zero role for this ADB-S database
- OSN: never enabled for this ADB-S — no listener active — zero internet exposure
- VCN DIVERSA via own SGW: not applicable — no public endpoint active on OSN — all paths are N/A
- Private DNS Zone: auto-provisioned by Oracle — both <adb-prefix>.oraclecloud.com and <adb-prefix>.oraclecloudapps.com resolve to PE <adb-pe-ip> within VCN
- HTTPS tools: not used in current architecture — not active

6.2 Configuration and Motivation

OCI Autonomous Database Serverless provisioned with "Private endpoint access only" in the VCN private subnet (<subnet-cidr>), PE private IP <adb-pe-ip>. Allow Public Access disabled. No public endpoint ever initialized on OSN. No public URL active for this database on any protocol. Service Gateway not involved for ADB-S. Zero internet exposure.

ADB-S HTTPS tools — APEX, ORDS, Database Actions, SQL Developer Web — are not used in the current architecture. The only traffic toward ADB-S is SQL*Net TCPS 1521 / mTLS 1522 directly to the PE private IP. With no HTTPS traffic toward ADB-S tools, there is no need to enable public access, involve the Service Gateway, set up a LB L7, or deploy a Network Firewall. The PE ONLY choice reflects the real operational model: SQL*Net via private PE, minimal architecture, zero attack surface.

6.3 SQL*Net — TCPS 1521 / mTLS 1522

Accessible exclusively via PE private IP (<adb-pe-ip>) from the VCN subnet. For on-premise workloads and corporate network access: via FastConnect or IPSec VPN through DRG toward the PE VCN, with hosts file configured to resolve ADB hostnames to the PE private IP (no SGW, no DRG for HTTPS/TCPS — direct access). Connectivity has been verified in the field also from local PC via VPN confirming correct operation on both TCPS and mTLS protocols.

6.4 HTTPS Port 443 — ADB-S Tools: Not Used

ADB-S HTTPS tools are not active in the current architecture. The URL <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com is technically reachable via private VCN DNS toward PE private IP (<adb-pe-ip>) but is not used. No transit on OSN. No SGW involved.

6.5 Service Gateway — Not Involved

The public endpoint was never enabled on OSN for this database. The SGW remains configurable in the VCN for other OSN services (Object Storage, etc.) but has no role in reaching ADB-S. No path via SGW toward this database exists or can be established.

6.6 Private DNS

Oracle automatically provisions private DNS zones in the VCN Resolver at PE deployment: <adb-prefix>.adb.eu-<region>-1.oraclecloud.com → <adb-pe-ip> and <adb-prefix>.adb.eu-<region>-1.oraclecloudapps.com → <adb-pe-ip>. For on-premise connections via DRG, conditional forwarding must be configured toward the OCI resolver exclusively for the specific database DNS zones, without forwarding the entire public Oracle zones in their entirety.

6.7 ZPR — Zero Trust Packet Routing

In this PE ONLY scenario, ZPR represents the most advanced security level applicable to the ADB-S PE. ZPR protects sensitive data from unauthorized access via intent-based policies written for OCI resources assigned security attributes. ZPR enforces policies at the network level for every access request, independently of potential modifications or misconfigurations of the network architecture. The key advantage in this scenario is that ZPR decouples the security policy from the network architecture: even if NSG or Security List were erroneously modified, ZPR policies continue to block any traffic not explicitly authorized toward PE <adb-pe-ip>. ZPR is offered at no additional cost for OCI.

```
# ZPR policy — PE ONLY, SQL*Net from VCN workload:
in networks:<your-vcn-name> VCN
allow app:<authorized-workload> endpoints
to connect to db:<adb-s-name> endpoints
with protocol='tcp/1522'

# For on-premise traffic via DRG (IP range):
in networks:<your-vcn-name> VCN
allow '<subnet-cidr>'
to connect to db:<adb-s-name> endpoints
with protocol='tcp/1522'
```

6.8 Future Evolution

Should the use of ADB-S HTTPS tools be required in the future, Scenario A must be adopted: public access enabled, ACL limited to LB L7 frontend IP only (<lb-frontend-ip>), Layer 7 Load Balancer, SGW, OCI Network Firewall with TLS inspection. This will ensure visibility of the real client IP at the Network Firewall log level, structurally compensating the Unified Audit Trail limitation due to the proxy nature of ORDS.

6.9 Security Implications

Zero public attack surface. No public endpoint on OSN. SQL*Net access bound exclusively to PE private IP (<adb-pe-ip>). ZPR as additional intent-based protection layer independent of network architecture. Fully compatible with IAM Database Token authentication, mTLS encryption in transit, Unified Audit Trail for complete SQL session tracking, and OCI Audit Logs for all Control Plane operations. Architecture compliant with enterprise security requirements and Zero Trust principles for data access.

7. Logging and Auditing

7.1 Separation of Logging Planes: Control Plane vs Data Plane

In ADB-S logging follows a strict separation between Control Plane and Data Plane. All service management operations — provisioning, scaling, network configuration, lifecycle, enable/disable public access — are tracked exclusively in OCI Audit Logs, immutable and centralized, queryable via OCI Logging. SQL activities and database access events never transit in OCI Audit logs but are tracked internally in the database via Unified Auditing Trail.

https://docs.oracle.com/en-us/iaas/Content/Audit/Concepts/auditoverview.htm

7.2 Control Plane Auditing — OCI Audit Logs

In Autonomous Database Serverless the Control Plane exclusively includes operations performed via OCI infrastructure APIs acting on the database resource as a cloud entity. These operations are processed outside the database, before any involvement of the Oracle Database kernel, and include REST calls made via OCI Console, OCI CLI, SDK, or direct APIs for: provisioning, network configuration update, endpoint management, scaling, backup, restore, and lifecycle management. Every API invocation is automatically tracked by OCI Audit Logs, which records operation metadata (caller, resource OCID, action, outcome and timestamp). OCI Audit Logs are tenancy-wide, immutable, and contain no SQL session information or application data, operating exclusively on the Control Plane.

https://docs.oracle.com/en-us/iaas/Content/Audit/Concepts/auditoverview.htm

7.3 Formal Separation Between Control Plane and Data Plane Auditing

Oracle ADB-S implements a formal and non-permeable separation between Control Plane and Data Plane, reflected directly in the respective auditing mechanisms. All service management operations are executed outside the database and tracked exclusively via OCI Audit Logs. OCI Audit Logs record infrastructure metadata: caller, principal OCID, resource OCID, operation, request parameters, response status, and timestamp. These logs are tenant-wide, immutable, and contain no SQL session information, database identity, or application content. Conversely, all activities occurring inside the database engine — login, privilege use, SQL statement execution, DDL, DML, and administrative database operations — are completely invisible to OCI Audit Logs and tracked only in the Unified Audit Trail.

https://docs.oracle.com/en-us/iaas/Content/Audit/Concepts/auditoverview.htm
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/adb-audit.html

7.4 Internal Architecture of the Unified Audit Trail

In ADB-S, Data Plane auditing is implemented exclusively via Unified Auditing, a native subsystem of the Oracle Database kernel. The Unified Audit Trail is not an external service nor an application layer, but an internal pipeline to the SQL engine, activated during authentication, authorization, and operation execution phases. Audit events are generated synchronously with the security event but persisted through internal database buffering mechanisms, designed to reduce the impact on application transactions. Audit records are materialized in the logical Unified Audit Trail, exposed via system views (primarily UNIFIED_AUDIT_TRAIL). In Autonomous Database direct access to underlying physical structures is intentionally precluded, guaranteeing operational immutability: records are not alterable or directly deletable by users, including database administrators. This constraint is an integral part of the managed service security model and ensures the forensic validity of audit data. The audit trail captures exclusively Data Plane events — those occurring after acceptance of the SQL*Net connection by the listener. No operations performed via OCI APIs (provisioning, scaling, lifecycle, network configuration) produce records in the Unified Audit Trail.

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/adb-audit.html
https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/AUDIT-Unified-Auditing.html

7.5 Audit Policy — Event Emission Decision Framework

Audit Policies define the semantic perimeter of events to audit but do not affect the informational structure of the audit record. From a formal standpoint, an audit policy is a set of rules evaluated by the Unified Auditing engine during event processing (statement parsing, privilege check, execution). If a policy is applicable the kernel emits an audit record; otherwise the event is ignored. Policies operate on well-defined categories (statement, privilege, action) and do not introduce or remove informational attributes from the trail. This model ensures that event selection logic is orthogonal to audit data representation logic.

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/adb-audit.html

https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/AUDIT-Unified-Auditing.html

7.6 Default Audit Policies on ADB-S — Active at Provisioning

In Autonomous Database Serverless, Oracle automatically pre-enables at provisioning a set of system audit policies covering critical security operations without requiring any action from the ADMIN user. The default Oracle policies active on ADB-S at provisioning include:

- ORA_LOGON_FAILURES — audits exclusively failed login attempts (ACTIONS LOGON WHENEVER NOT SUCCESSFUL)
- ORA_DATABASE_PARAMETER — audits modifications to database parameters via ALTER SYSTEM
- ORA_SECURECONFIG — audits security configuration operations including user, role, privilege, and audit policy management
- ORA_ACCOUNT_MGMT — audits creation, modification, and deletion of database accounts
- ORA_DBA_AUDITING — audits actions of users with DBA privileges
- ORA_CIS_RECOMMENDATIONS — implements CIS Oracle Database Benchmark recommendations for compliance auditing

WARNING *The ADMIN user can add custom policies or enable additional Oracle predefined policies via standard Unified Auditing DDL commands, but CANNOT disable or modify Oracle predefined system policies — these are immutable and an integral part of the managed service security model.*

https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/adb-audit.html

7.7 Unified Audit Trail Purge — Native 14-Day Window and Manual Purge

In Autonomous Database Serverless, Oracle runs an automatic daily job that deletes Unified Audit Trail records with EVENT_TIMESTAMP older than 14 days. This window is fixed, non-configurable, and operates independently of any other setting. It is the critical pressure point of the entire retention architecture: if Data Safe does not collect records within 14 days of their generation, they are irreversibly lost. Records older than 14 days are automatically deleted independently of any other configuration, unless Data Safe has updated the LAST_ARCHIVE_TIMESTAMP to protect records not yet archived.

To perform a manual early purge — for example to free space or delete records already archived on Data Safe — two procedures must be executed in sequence by a user with the AUDIT_ADMIN role. The first procedure sets the reference timestamp:

```
BEGIN
  DBMS_AUDIT_MGMT.SET_LAST_ARCHIVE_TIMESTAMP(
    AUDIT_TRAIL_TYPE  => DBMS_AUDIT_MGMT.AUDIT_TRAIL_UNIFIED,
    LAST_ARCHIVE_TIME => SYSTIMESTAMP - INTERVAL '7' DAY
  );
END;
/
```

The second procedure executes the actual purge of all records older than the set timestamp:

```
BEGIN
  DBMS_AUDIT_MGMT.CLEAN_AUDIT_TRAIL(
    AUDIT_TRAIL_TYPE        => DBMS_AUDIT_MGMT.AUDIT_TRAIL_UNIFIED,
    USE_LAST_ARCH_TIMESTAMP => TRUE
  );
END;
/
```

WARNING *SET_LAST_ARCHIVE_TIMESTAMP must always precede CLEAN_AUDIT_TRAIL. Executing CLEAN_AUDIT_TRAIL without having set the timestamp produces an error. In environments integrated with Data Safe, do not perform manual purges independently of the Data Safe collection cycle.*

https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/adb-audit.html

7.8 Unified Audit Trail Enrichment — AUDIT CONTEXT and APPLICATION_CONTEXTS

The contextual enrichment mechanism allows capturing additional SYS_CONTEXT attributes in the APPLICATION_CONTEXTS field of each audit record, providing identity and connection information that standard audit policies do not include by default. This is particularly relevant in environments using IAM Token authentication, where the real caller identity is expressed via USERENV namespace attributes of SYS_CONTEXT. The command to enable enrichment is AUDIT CONTEXT, executable by the ADMIN user with AUDIT_ADMIN role. The most relevant attributes for the IAM Token authentication scenario are:

- IDENTIFICATION_TYPE — GLOBAL for users mapped to IAM identity, LOCAL for local database users with password — fundamental discriminant for distinguishing IAM sessions from local sessions

- GLOBAL_UID — contains the OCID of the IAM user authenticated via token, providing direct correlation with the OCI identity
- AUTHENTICATION_TYPE — indicates the authentication mechanism used (PASSWORD, KERBEROS, or IAM type for token)
- CLIENT_IDENTIFIER — can be set by the application via DBMS_SESSION.SET_IDENTIFIER to propagate the real application end-user identity in connection pool architectures
- IP_ADDRESS — reports the IP of the TCP connection to the listener — useful for direct SQL*Net connections, but subject to the structural ORDS proxy limitation for HTTPS sessions

Once configured, the APPLICATION_CONTEXTS field contains values in structured format NAMESPACE=USERENV,ATTRIBUTE=<name>,VALUE=<value> for each enabled attribute.

7.9 SYS_CONTEXT — Formal Model of the Session Context

SYS_CONTEXT represents the mechanism through which the Oracle Database kernel exposes the contextual state of the session. The USERENV namespace is populated internally by the database during authentication and session management phases and contains attributes describing the caller identity, connection origin, and execution context. These attributes are not derived from application inputs nor modifiable by the user, constituting an authoritative and non-falsifiable source for security event correlation. SYS_CONTEXT represents the convergence point between database identity, external IAM identity, and real connection parameters.

https://docs.oracle.com/en/database/oracle/oracle-database/26/sqlrf/SYS_CONTEXT.html

7.10 Context Auditing and Unified Audit Trail Enrichment

Context auditing is a mechanism distinct from audit policies that allows specifying which SYS_CONTEXT attributes must be captured in audit records. When configured, the database serializes the values of selected attributes in the APPLICATION_CONTEXTS field of the Unified Audit Trail. This field is a structured container that deterministically associates namespace, attribute, and value to the single audited event. Enrichment occurs at record generation time, ensuring temporal and semantic consistency between event and context. The result is an audit trail in which each record represents a complete snapshot of the session security state at the time of the event.

https://docs.oracle.com/en/database/oracle/oracle-database/26/dbseg/creating-custom-unified-audit-policies.html

7.11 Data Safe as Data Plane Audit Aggregation Layer

Oracle Data Safe uses the Unified Audit Trail as the primary source for Database Activity Monitoring, compliance reporting, and centralized audit data-plane storage. Data Safe does not replace the database auditing mechanism but orchestrates its collection, archiving, and lifecycle management (including auto-purge and archiving based on LAST_ARCHIVE_TIMESTAMP). The Data Safe repository is conceptually separate from the source database and allows managing retention policies, billing, and reporting at scale.

https://docs.oracle.com/en/cloud/paas/data-safe/udscs/audit-trails.html

7.12 Pipeline to OCI Logging — Supported Models

For centralizing data-plane logs in OCI Logging, Oracle documents two distinct models. The first model — suitable for limited contexts — involves extracting data from Data Safe and direct ingestion into OCI Logging via OCI Functions. This approach leverages the centralized Data Safe repository but scales less effectively in high-density multi-database environments.

https://docs.oracle.com/en/learn/datasafe-audit-log-to-oci-logging/index.html
https://github.com/oracle-devrel/technology-engineering/tree/main/security/security-design/shared-assets/fn-datasafe-dbaudit-to-oci-logging

The second model — recommended for enterprise scenarios — uses per-database pipelines: each ADB-S exports increments of the Unified Audit Trail toward Object Storage via Oracle-maintained pipelines or PL/SQL (DBMS_CLOUD.EXPORT_DATA). From there, a centralized Function performs ingestion toward OCI Logging. This model preserves per-database separation, improves scalability, and reduces contention on the Data Safe repository.

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/pipeline-oracle-maintained.html

8. Oracle Data Safe — ADB-S Private Endpoint — Complete Guide

8.1 IAM Prerequisites — Necessary and Sufficient Policies

To operate with Data Safe on ADB-S with Private Endpoint three distinct policy blocks are required, reflecting three separate OCI resources.

Block 1 — ADB-S Database Operations

Granular least privilege with any-of condition limited to the three necessary operations:

```
allow group <group-name> to use autonomous-databases in compartment <compartment-name>
where any {
  request.operation = 'GetAutonomousDatabase',
  request.operation = 'RegisterAutonomousDatabaseDataSafe',
  request.operation = 'DeregisterAutonomousDatabaseDataSafe'
}
```

Block 2 — Target Database Management in the Data Safe Repository

The compartment of the Data Safe target does not necessarily need to coincide with that of the ADB-S instance. Oracle Data Safe automatically registers the target in the same compartment where the database resides, but the policy on target-databases can point to a dedicated Data Safe compartment.

```
allow group <group-name> to manage target-databases in compartment <compartment-name>
```

https://docs.oracle.com/en-us/iaas/data-safe/doc/target-database-registration-overview.html

Block 3 — Data Safe Private Endpoint (Mandatory for ADB-S with PE ONLY)

```
-- New Data Safe PE creation:
allow group <group-name> to manage data-safe-private-endpoints in compartment <compartment-name>
allow group <group-name> to manage virtual-network-family in compartment <compartment-name>

-- Reuse of existing Data Safe PE (least privilege):
allow group <group-name> to use data-safe-private-endpoints in compartment <compartment-name>
allow group <group-name> to use virtual-network-family in compartment <compartment-name>

-- Maximum network granularity (VNIC + subnet + NSG):
allow group <group-name> to manage vnics in compartment <compartment-name>
allow group <group-name> to use subnets in compartment <compartment-name>
-- Only if NSG are used on the Data Safe PE:
allow group <group-name> to use network-security-groups in compartment <compartment-name>
```

https://docs.oracle.com/en/cloud/paas/data-safe/admds/permissions-register-autonomous-database.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/autonomous-database-iam-policies.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/create-oracle-data-safe-private-endpoint.html
https://docs.oracle.com/en/cloud/paas/data-safe/dsiad/datasafe_private_endpoint.html

8.2 Data Safe Private Endpoint — Network Requirements

The Data Safe Private Endpoint materializes as a VNIC with a private IP in a subnet of choice in the VCN. The network requirement is not co-location on the same subnet as the ADB-S PE: it is sufficient that routing exists between the two subnets and that Security List or NSG allow inbound traffic on port 1522 (mTLS) toward the ADB-S PE (<adb-pe-ip>). A maximum of one Data Safe Private Endpoint per VCN is allowed — if one already exists in the same VCN it is reused for all target databases in that VCN.

8.3 Complete Least Privilege Policy — PE Only Scenario Synoptic

```
-- [1] Database — least privilege only Data Safe operations
allow group DataSafe-Operators to use autonomous-database
  in compartment <adb-compartment>
  where any {
    request.operation = 'GetAutonomousDatabase',
    request.operation = 'RegisterAutonomousDatabaseDataSafe',
    request.operation = 'DeregisterAutonomousDatabaseDataSafe'
  }
```

```
-- [2] Target management in Data Safe repository
allow group DataSafe-Operators to manage target-databases
  in compartment <datasafe-compartment>
-- [3a] Data Safe PE — reuse existing
allow group DataSafe-Operators to use data-safe-private-endpoints
  in compartment <network-compartment>
-- [3b] Network granular minimum (VNIC + subnet + optional NSG)
allow group DataSafe-Operators to manage vnics
  in compartment <network-compartment>
allow group DataSafe-Operators to use subnets
  in compartment <network-compartment>
allow group DataSafe-Operators to use network-security-groups
  in compartment <network-compartment>
```

https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/autonomous-database-iam-policies.html

https://docs.oracle.com/en-us/iaas/data-safe/doc/register-autonomous-databases-shared-exadata-infrastructure-private-vcn-access.html

8.4 ADB-S Target Registration — Operational Procedure

Registration can occur in two equivalent ways: (1) from the ADB-S instance detail page in the OCI Console, section "Oracle Data Safe" → "Register" button — fastest for individual databases; (2) from the Data Safe console: Data Safe → Target Databases → Register Database → select "Oracle Autonomous Database" → insert the database OCID, compartment, select the existing Data Safe Private Endpoint. In both cases, upon registration completion Oracle automatically unlocks the pre-seeded DS$ADMIN account and starts the discovery process for available audit trails. Upon deregistration, data already collected in the Data Safe repository is preserved according to the retention configured before deregistration — not deleted immediately.

https://docs.oracle.com/en/cloud/paas/data-safe/admds/create-iam-policies-oracle-data-safe-users.html

8.5 DS$ADMIN — Default Roles and Additional Role Enablement

At ADB-S Serverless target registration, Oracle automatically assigns to DS$ADMIN: DS$ASSESSMENT_ROLE (default) and DS$AUDIT_COLLECTION_ROLE (default). All other roles require explicit enablement:

```
-- Enable Audit Policy Management (required to manage policies from Data Safe console):
EXEC DS_TARGET_UTIL.GRANT_ROLE('DS$AUDIT_SETTING_ROLE');
-- Enable Data Discovery:
EXEC DS_TARGET_UTIL.GRANT_ROLE('DS$DATA_DISCOVERY_ROLE');
-- Enable Data Masking (also requires UNLIMITED TABLESPACE):
EXEC DS_TARGET_UTIL.GRANT_ROLE('DS$DATA_MASKING_ROLE');
GRANT UNLIMITED TABLESPACE TO DS$ADMIN;
-- Enable SQL Firewall Management:
EXEC DS_TARGET_UTIL.GRANT_ROLE('DS$SQL_FIREWALL_ROLE');
-- Revoke a role if not needed:
EXEC DS_TARGET_UTIL.REVOKE_ROLE('DS$DATA_MASKING_ROLE');
```

https://docs.oracle.com/en-us/iaas/data-safe/doc/roles-oracle-data-safe-service-account.html

https://docs.oracle.com/en-us/iaas/data-safe/doc/grant-and-revoke-roles-oracle-data-safe-service-your-target-database.html

8.6 Features Available on ADB-S Target in Data Safe

Activity Auditing

Continuous collection of Unified Audit Trail from the database into the centralized Data Safe repository. Retention up to 12 months online and up to 7 years in archive — extends the native ADB-S 14-day window. Includes lifecycle management of records, coordination of purge via LAST_ARCHIVE_TIMESTAMP, and free threshold of 1 million records/month per target.

Audit Policy Management

Management of audit policies directly from the Data Safe console without executing DDL on the database (requires DS$AUDIT_SETTING_ROLE). Oracle Data Safe predefined policies available: failed accesses, ADMIN activity, schema changes, user changes, activity on sensitive data. CIS and STIG policy sets selectable as preconfigured sets. Custom policies existing on the database are visible and manageable from the console.

Alerts — Operationally Relevant Examples

- Login failed ADMIN user — alert on every failed access attempt of the local ADMIN user

- Local user login — alert when a database user authenticates with local password instead of IAM Token, unexpected behavior in IAM-only architecture
- GRANT of critical privileges — alert on GRANT of DBA, SYSDBA, EXEMPT ACCESS POLICY, SELECT ANY DICTIONARY or critical application roles
- Full table scan on sensitive tables — alert on SELECT without WHERE filter on tables classified as sensitive by Data Discovery (requires DS$DATA_DISCOVERY_ROLE for classification)
- Audit policy modification — alert when an existing audit policy is modified or disabled
- Database user creation or deletion — alert on CREATE USER / DROP USER executed from any session
- Database parameter modification — alert on ALTER SYSTEM SET

NOTE *Alerts are visible in the Data Safe dashboard and integrable with OCI Notifications for notification via email, Slack, PagerDuty, or webhook.*

Security Assessment

Periodic analysis of database configuration against Oracle, CIS, and STIG best practices. Findings categorized by risk level (High / Medium / Low / Advisory). Supports baseline definition and drift monitoring over time.

User Assessment

Analysis of all database accounts: identifies high-risk users by privilege profile, inactive accounts, accounts with default passwords, accounts with unnecessary administrative privileges. Baseline and drift monitoring over time.

Data Discovery

Schema scan to identify columns containing sensitive data (PII, financial data, credentials). Results used by Activity Auditing for alerts on sensitive data access and by Data Masking for masking.

Data Masking

Irreversible or format-preserving masking of sensitive data for non-production environments. Requires explicit DS$DATA_MASKING_ROLE enablement via DS_TARGET_UTIL.

SQL Firewall

Control of authorized SQL statements per database user via allowlist generated from a training period. Blocks or logs SQL statements not conforming to the expected profile. Requires explicit DS$SQL_FIREWALL_ROLE enablement via DS_TARGET_UTIL.

8.7 IAM Policy for Data Safe — Activity Auditing, Audit Trail and Alerts

For operational Activity Auditing and Alerts work, the reference resource families are: data-safe-audit-family for audit trail collection, audit policy management, and audit reports; data-safe-alert-family for alert policy management, target-policy associations, and generated alerts. The data-safe-family aggregate includes all six families (assessment, audit, discovery, masking, alert, sql-firewall).

```
-- Full admin access to all Data Safe features:
allow group DataSafe-Admins to manage data-safe-family in compartment <compartment-name>

-- Operational group — Activity Auditing and Alerts only (least privilege):
allow group DataSafe-AuditOps to manage data-safe-audit-family in compartment <compartment-
name>
allow group DataSafe-AuditOps to manage data-safe-alert-family in compartment <compartment-
name>

-- Read-only group — audit reports and alerts consultation only:
allow group DataSafe-Readers to read data-safe-audit-family in compartment <compartment-
name>
allow group DataSafe-Readers to read data-safe-alert-family in compartment <compartment-
name>

-- Security ops — assessment + read audit/alerts, no registration/masking/discovery:
allow group DataSafe-SecurityOps to manage data-safe-assessment-family in compartment
<compartment-name>
allow group DataSafe-SecurityOps to read data-safe-report-definitions in compartment
<compartment-name>
allow group DataSafe-SecurityOps to read data-safe-reports in compartment <compartment-
name>
allow group DataSafe-SecurityOps to read data-safe-alerts in compartment <compartment-name>
```

https://docs.oracle.com/en/cloud/paas/data-safe/admds/create-iam-policies-oracle-data-safe-users.html

https://docs.oracle.com/en/cloud/paas/data-safe/dsiad/datasafe_iam.html

8.8 Granular IAM Policy per Resource-Type

For users who must operate on Activity Auditing, Alert and Audit Policy without access to Data Discovery, Masking and SQL Firewall:

```
-- Audit trail management and log collection:
allow group DataSafe-Operators to manage data-safe-audit-trails in compartment
<compartment-name>
-- Audit profile management (retention, paid usage):
allow group DataSafe-Operators to manage data-safe-audit-profiles in compartment
<compartment-name>
-- Read collected audit events:
allow group DataSafe-Operators to read data-safe-audit-events in compartment <compartment-
name>
-- Audit policy management on target:
allow group DataSafe-Operators to manage data-safe-audit-policies in compartment
<compartment-name>
-- Alert policy management and target-alert associations:
allow group DataSafe-Operators to manage data-safe-alert-policies in compartment
<compartment-name>
allow group DataSafe-Operators to manage data-safe-target-alert-policy-associations in
compartment <compartment-name>
-- Generated alerts:
allow group DataSafe-Operators to manage data-safe-alerts in compartment <compartment-name>
-- Reports (read only):
allow group DataSafe-Operators to read data-safe-reports in compartment <compartment-name>
allow group DataSafe-Operators to read data-safe-report-definitions in compartment
<compartment-name>
```

For all features by family (simpler):

```
allow group DataSafe-Operators to manage data-safe-audit-family in compartment
<compartment-name>
allow group DataSafe-Operators to manage data-safe-alert-family in compartment
<compartment-name>
allow group DataSafe-Operators to read data-safe-report-definitions in compartment
<compartment-name>
allow group DataSafe-Operators to read data-safe-reports in compartment <compartment-name>
-- Full admin:
allow group Data-Safe-Admins to manage data-safe-family in tenancy
```

https://docs.oracle.com/en/cloud/paas/data-safe/admds/data-safe-family-resource.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/create-iam-policies-oracle-data-safe-users.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/autonomous-database-iam-policies.html

8.9 Activity Auditing — How to Work with Logs and the Audit Trail

Once the ADB-S target is registered and the Audit Trail started, Data Safe continuously collects records from the database Unified Audit Trail and copies them to its centralized repository. The operational entry point is Security Center — Activity Auditing in the OCI console, giving access to: Audit Trails, Audit Profiles, Audit Policies, All Activity Reports, and Audit Insights.

The Audit Trail for ADB-S is of type UNIFIED_AUDIT_TRAIL and must be started explicitly the first time: Security Center → Activity Auditing → Related Resources → Audit Trails → select target → Start. Collection is incremental: Data Safe reads only new records compared to the last completed cycle, updating the LAST_ARCHIVE_TIMESTAMP in the source database at each cycle. This timestamp is the coordination mechanism between the native 14-day window of ADB-S and Data Safe archiving: as long as Data Safe is collecting and updating this timestamp, the automatic ADB-S purge does not delete records not yet archived.

The Audit Profile governs retention settings and collection behavior for each target: online retention period (default 12 months, maximum 12 months); archive period (0 to 72 additional months for total retention up to 7 years). The option to override global retention settings allows differentiated configurations per target. Retention modification applies to new incoming data immediately, without retroactive effect.

8.10 All Activity Reports — Predefined and Custom Reports

Data Safe offers predefined out-of-the-box reports: Login Activity (all database accesses with outcome, user, client host, timestamp); Failed Logins (exclusively failed attempts); Admin Activity (all actions of administrative users); User Changes (creation, modification, deletion of users and roles); Audit Policy Changes (modifications to audit policies); Database Parameter Changes (ALTER SYSTEM SET); Data Modification (DML on sensitive

tables). Each report is filterable by target database, user, time period, action type, client host, and other Unified Audit Trail attributes. Reports can be exported in PDF or XLS. Custom Reports can be saved with filters and customizations and scheduled for periodic automatic generation with distribution via OCI Notifications. The Audit Insights feature allows analyzing audit volume per target, schema, database user, operation type, and client — useful for identifying anomalies in access patterns.

https://docs.oracle.com/en/cloud/paas/data-safe/udscs/alerts-overview.html
https://docs.oracle.com/en/learn/oracle-datasafe-db-security-mgmt/index.html

8.11 Alerts — Predefined Policies, Target Associations, OCI Notifications

The Data Safe alerting system is based on alert policies evaluated continuously against collected audit records. The prerequisite chain is always: active audit policy on DB → event generated in Unified Audit Trail → collected by Data Safe → evaluated by alert policy → alert emitted if it matches. Predefined alert policies available for ADB-S cover: database parameter modifications; audit policy modifications; failed logins from admin user; user entitlement modifications (GRANT/REVOKE); database user creation and deletion; user profile modifications. Each predefined policy has an associated severity level — Critical, High, Medium, or Low.

The association between alert policy and target database is managed from Target-Policy Associations in Security Center — Alerts. Disabling an alert policy suspends evaluation but does not interrupt audit data collection — useful during maintenance windows. Custom alert policies allow monitoring events not covered by predefined policies: parameter modifications from non-allowlisted client hosts; accesses to important tables from unauthorized users; stored procedure definition modifications; Database Vault realm violations. Default alert retention is 3 months in the Data Safe repository.

For proactive notification: subscribe to the Alert Generated event or configure an OCI Monitoring alarm. Data Safe allows creating notifications directly from its console. Supported notification channels via OCI Notifications: email, SMS, Slack, webhook toward REST endpoints, and OCI Functions calls. For the enterprise standard scenario: subscribe at minimum to Alert Generated with filter on severity Critical and High toward an email channel or corporate SIEM via webhook, and configure a separate alarm for anomalous alert volume (e.g. more than 1000 alerts in 5 minutes).

https://docs.oracle.com/en/cloud/paas/data-safe/udscs/alerts-overview.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/event-types-oracle-data-safe.html
https://blogs.oracle.com/database/data-safe-notifications

8.12 Activity Auditing — Audit Trail: Start, Operational States and Management

Collection does NOT start automatically after registration: the trail remains in NOT_STARTED state until explicitly started. Path: Security Center → Activity Auditing → Related Resources → Audit Trails → select target → Start → select collection start date → confirm. Before starting collection verify the estimated volume in the "To Be Collected" column. Operational states of the trail:

- NOT_STARTED — trail created at registration but not yet started
- STARTING — during the first startup phase
- COLLECTING — trail actively collecting records from the Unified Audit Trail
- IDLE — trail active and functional but no new records to collect at this moment
- RECOVERING — trail encountered an error, attempting to return to COLLECTING, reprocessing from last checkpoint to avoid duplicates
- RESUMING — trail resuming collection after a stop
- RETRYING — during the attempt to enter RESUMING state
- STOPPING — during manual stop
- STOPPED — trail manually stopped
- STOPPED_NEEDS_ATTN — after 4 hours of failed retries following ADB-S stop/restart, trail requires manual resume from operator
- STOPPED_FAILED — target database deleted

WARNING *In case of stop or restart of the ADB-S instance, the trail automatically enters retry state for four hours. If after four hours the database is unavailable, the trail enters STOPPED_NEEDS_ATTN and requires a manual resume.*

https://docs.oracle.com/en-us/iaas/data-safe/doc/view-and-manage-audit-trails.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/adb-audit.html

8.13 Audit Profile — Retention and Paid Usage

Configurable in Data Safe → Security Center → Activity Auditing → Audit Profiles → select target. Configurable parameters per target: online retention period (1 to 12 months); behavior at reaching the monthly million

(continue/stop); auto-purge enable/disable. Modification applies to new incoming data immediately, without retroactive effect.

https://docs.oracle.com/en/cloud/paas/data-safe/udscs/using-oracle-data-safe.pdf

8.14 Audit Policy on ADB-S from Data Safe — Provisioning, Enablement, Customization

Data Safe allows managing audit policies on the ADB-S target directly from the console, provided DS$ADMIN has DS$AUDIT_SETTING_ROLE assigned. Flow: Data Safe → Security Center → Activity Auditing → Audit Policies → select target. Available policies: predefined Oracle Data Safe policies, predefined Oracle Database policies, individual custom policies, CIS and STIG compliance policies. For some policies it is possible to specify the users to audit or exclude. Custom policies created directly on the database via DDL are visible in the "Additional Policies" section and are enableable/disableable from the console without direct SQL access.

https://docs.oracle.com/en/cloud/paas/data-safe/admds/data-safe-family-resource.html
https://docs.oracle.com/en/cloud/paas/data-safe/udscs/using-oracle-data-safe.pdf

9. Data Safe — Audit Data Retention: Online, Archive, Purge and ADB-S Coordination

9.1 Architecture Overview — Three Distinct Strata

Retention management of audit data in Oracle Data Safe is one of the most critical architectural aspects to plan correctly before starting collection, because configurations apply to new incoming data and not retroactively to already archived data. Data Safe clearly separates the audit data lifecycle into three distinct strata: the native window of the source database (ADB-S); the online storage of the Data Safe repository; and the long-term offline archive.

9.2 Stratum 1 — Native ADB-S Window: 14 Days, Automatic, Non-Configurable

Oracle automatically runs a daily job that deletes from the UNIFIED_AUDIT_TRAIL all records with EVENT_TIMESTAMP older than 14 days. This window is fixed, non-configurable, and operates independently of any other setting. It is the critical pressure point of the entire retention architecture: if Data Safe does not collect records within 14 days, they are irreversibly lost. The coordination mechanism is the LAST_ARCHIVE_TIMESTAMP (LAT): every time Data Safe completes a collection cycle it updates the LAT in the source database, signaling that records up to that timestamp have been safely archived and can be deleted from the local window.

9.3 Stratum 2 — Online Data Safe Storage: 1 to 12 Months

An audit profile defines the online retention period, offline retention period, and paid usage settings for a target database. Online retention is configurable from a minimum of 1 month to a maximum of 12 months. Records within the online window are directly queryable from Data Safe reports without requiring retrieval operations. Configuration is in Data Safe → Security Center → Activity Auditing → Audit Profiles → select target → Update Retention. The option "Do you want to override the global retention settings?" allows differentiated configurations per target database with different compliance requirements. Retention modification applies to new incoming data immediately, without retroactive effect on already archived records.

9.4 Stratum 3 — Offline Storage: 0 to 72 Months, Total Maximum 7 Years

The offline retention period specifies the number of months to retain audit data in offline storage. Minimum configurable is zero months, maximum is 72 months (six years). Total possible retention is seven years: one year online and six years in archive. Records in archive are not directly queryable: they must first be retrieved via archive retrieval operation. Maximum six retrieves per month per target database; if operational volume requires more, a Service Request to Oracle Support is required specifying the additional retrieves needed — the increased limit applies to all target databases in the tenancy. If regulatory requirements require retention beyond 7 total years, Oracle Support must be contacted.

9.5 Monthly Volume — 1 Million Free Records, Unlimited at Cost

Oracle Data Safe can collect an unlimited number of audit records per month per target database. The first million records per month per target database is free. Records generated by DS$ADMIN are NOT counted toward the monthly limit. Upon exceeding the monthly million the default behavior is to continue collection at additional cost per record. Configurable both globally from Data Safe → Settings → Global Settings, and per individual target via the Audit Profile.

The "Compute Audit Volume" section in the Audit Profile allows calculating: volume of records available on the target not yet collected; volume of records already collected available for online reporting; and volume of archived offline records. If a target database is deregistered, audit data already collected is preserved according to the online and offline retention periods configured before deregistration.

9.6 Auto-Purge — Critical Coordination with the Native ADB-S 14-Day Window

The Data Safe auto-purge feature allows deleting audit records from the target database on a regular basis. When auto-purge is enabled, a daily job verifies the LAT and deletes all audit records that are seven days older than the LAT value. Auto-purge is disabled by default and must be enabled consciously. Enabling auto-purge deletes all audit records in the database trail every seven days, including those older than the initial collection start date. Records could be deleted even if not yet collected in Oracle Data Safe.

WARNING *Performing purges outside the Data Safe framework — such as directly using DBMS_AUDIT_MGMT.CLEAN_AUDIT_TRAIL — can delete records before they are collected, especially*

when auto-purge is disabled and LAT updating is deactivated. Even with auto-purge disabled in Data Safe, the native ADB-S job continues to delete records older than 14 days independently — the LAT updated by Data Safe is the only mechanism protecting uncollected records from the native database window.

10. Monitoring — Metrics, Alerts and Events Rules

10.1 Service Metrics — OCI Monitoring

Autonomous Database Serverless exposes an official set of Service Metrics usable to build OCI Alarms for operational purposes. This monitoring layer is continuous and quantitative, designed to control service state and performance — compute utilization, sessions, storage, and other operational indicators. The complete list of Oracle-supported metrics for ADB-S is available in official documentation and is the canonical reference for threshold-based alarm configuration.

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-monitor-metrics-list.html

> **NOTE** *Fundamental architectural note: ADB-S metrics are suitable for capacity/resource alerts (trends and thresholds) but do NOT provide a dedicated and reliable metric for detecting automatic managed backup fault (e.g., backup failed) as a direct failure process signal.*

10.2 Automatic Backups — Event-Driven via OCI Events Rules

Automatic backups in ADB-S are managed by the service — they are not a user-managed applicative job and are not exposed as a "failure counter" metric ready to alarm. For this reason the correct and supported method to detect execution and outcome of automatic backups is the event-driven approach via OCI Events Rules. Oracle exposes Database Service events related to the automatic backup lifecycle — specifically begin and end — interceptable in real time by creating rules on OCI Events.

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-database-events.html#GUID-8DBAD1C3-55BE-49F8-AB43-AB3EA708AF03

10.3 Backup Event — lifecycleState Field and Fault Condition

Backup completion is intercepted via event type: com.oraclecloud.databaseservice.automaticbackupautonomousdatabase.end. The backup outcome is discriminated via the lifecycleState field. From tests executed and notification received: when the automatic backup terminates correctly the payload reports "lifecycleState": "ACTIVE". To intercept fault conditions:

```
MATCH event WHERE (
  eventType EQUALS ANY OF (
    com.oraclecloud.databaseservice.automaticbackupautonomousdatabase.end
  )
  AND (lifecycleState MATCHES ANY OF (FAILED))
)
```

Support indicated as payload reference the JSON path data.lifecycleState. The rule is associated with the target compartment and, where necessary, further constrained to the ADB resource via OCID/resourceId. As Action it uses an OCI Notifications Topic for automatic alerts via email. The same backup events (begin/end) are also visible at OCI Audit level, ensuring traceability and correlation for operational/compliance purposes.

10.4 Alerting on Automatic Backups — Events Rule + Notifications

Backup-centric monitoring of automatic backups in Autonomous Database Serverless is implemented in event-driven mode via OCI Events Service. The solution consists of defining an Events Rule on the Database Service that intercepts the automatic backup end event. The rule evaluates the lifecycleState field of the event payload as the discriminant of operational outcome: correct completion → "lifecycleState": "ACTIVE"; fault → filter FAILED state in the condition. The action is an OCI Notifications Action toward a Topic with email subscription, obtaining near real-time alerts with "per-single-backup" granularity, avoiding indirect or non-deterministic approaches.

10.5 Auditability — Evidence and Traceability via OCI Audit

Automatic ADB-S backup execution is also traceable via OCI Audit, where events associated with the backup lifecycle (begin/end) are recorded with sufficient informational detail for operational correlation and ex-post verification: resource/OCID identifiers, timestamps, and event metadata. This enables an enterprise control model in which: (i) the notification from the Events Rule represents the immediate alerting mechanism; (ii) OCI Audit constitutes the objective and non-repudiable evidence source for troubleshooting, timeline reconstruction, and compliance/governance requirements.

For SOC and security control requirements, automatic backup execution of Autonomous Database Serverless is verifiable via OCI Audit, where events are recorded with complete informational payload. This means the SOC has an authoritative and immutable source (control-plane audit trail) to demonstrate that the automatic backup has started and terminated, correlating each event to: resource OCID, compartment, timestamp, event type, and

operational details. The Audit Log enables: evidence-based monitoring of actual automatic backup execution; correlation between notification received via OCI Events/Notifications and official audit record; incident response and post-mortem activities (root cause timeline reconstruction) in case of anomalies. In an enterprise governance model, this traceability enables integrating automatic backup logs into SOC processes (SIEM), enabling compliance controls and periodic verification of the continuity of managed data protection mechanisms.

10.6 Monitoring Architecture — Control Plane vs Data Plane Separation

OCI Autonomous Database Serverless exposes operational and performance metrics via OCI Monitoring, with automatic collection and without requiring agents or client-side configurations. Monitoring is formally separated between Control Plane (OCI) and Data Plane (Database Engine). The two components use different instrumentation, metrics, and pipelines and do not overlap.

11. Storage and Backup

11.1 Architecture Overview

Autonomous Database Serverless uses storage managed natively by Oracle, completely integrated in the service and not directly accessible to the customer. The backup is part of the OCI Control Plane and requires no planning or manual management. Primary data resides on ADB-S managed storage, while backups are stored on Oracle-managed cloud object storage — completely opaque to the customer: not visible as a bucket in the tenancy, not downloadable, not directly accessible.

11.2 Automatic Backups — Always Active, Non-Disableable, RPO < 10 Seconds

ADB-S automatically executes periodic backups including full, cumulative, and incremental (archived redo logs) to ensure data reliability and recoverability. Every instance is continuously safeguarded to achieve a Recovery Point Objective (RPO) of less than 10 seconds. Automatic backups are mandatorily activated during database creation and cannot be disabled — they are an integral and non-negotiable part of the managed service security model. Oracle internally manages scheduling, execution, and verification of backups without any applicative or DBA intervention. Automatic backups are retained even in case of database termination, until the end of the configured retention period.

11.3 Automatic Backup Retention — ECPU vs OCPU

Default automatic backup retention is 60 days. For ECPU compute model: retention is configurable from 1 to 60 days. The retention modification option is available exclusively for instances using the ECPU compute model. For OCPU compute model: retention is fixed at 60 days and not modifiable via console. The OCPU model was retired by Oracle for new provisioning from January 2024, therefore all new ADB-S databases use ECPU and benefit from configurable retention.

11.4 Backup Storage Billing — ECPU vs OCPU

Billing for automatic backups depends on the compute model: with OCPU, backup storage is included in the database storage cost; with ECPU, backup storage is billed separately and in addition to database storage. On ECPU the backup storage cost follows the OCI Object Storage model based on space actually occupied.

11.5 Point-in-Time Recovery (PITR)

PITR is supported across the entire configured retention period. It is possible to: restore the existing database to a specific point in time — an operation that overwrites the current database; or create a new Autonomous Database from a backup, preserving the original database. Both operations are executable via OCI Console, OCI CLI, REST API, or SDK. ADB-S automatically restores and recovers the database to the specified time point or using the backup selected from the list of available backups.

11.6 Backup Replication and Immutability

All backups managed by Oracle in ADB-S are immutable — they cannot be deleted or modified. In regions with multiple Availability Domains, backups are automatically replicated between ADs; in regions with a single AD, they are replicated between Fault Domains. This guarantees data durability and geographic resilience within the same OCI region. If the database is configured with Autonomous Data Guard, automatic backups are also replicated in the standby region to maximize resilience.

11.7 Immutable Backup Retention

The Immutable Backup Retention option locks the automatic backup retention period preventing modifications. When enabled, ADB-S does not allow changes to the retention period, preventing unauthorized removal of older backups by modifying retention. The option is configurable at provisioning or subsequently via modification of instance settings. Once enabled, immutable backup retention cannot be disabled nor the retention period modified via console or API: a Service Request to Oracle Cloud Support is required. Immutable backup retention applies exclusively to automatic backups and does not affect long-term backups.

11.8 Long-Term Backup — 3 Months to 10 Years

Via the long-term backup feature it is possible to create a backup as a one-time operation or as a scheduled long-term backup with retention from a minimum of 3 months to a maximum of 10 years. ADB-S automatically executes scheduled long-term backups according to the defined schedule: one-time, weekly, monthly, or yearly. Long-term

backups are standalone backups that allow creating a point-in-time clone based on the backup timestamp. It is NOT possible to use a long-term backup to recover or restore the same database from which the backup was created — long-term backups are usable exclusively to create a new database. The retention of an existing long-term backup is modifiable via console in the database Backups section, in the range from 3 months to 10 years. Long-term backups are billed as additional storage on both ECPU and OCPU models.

> **NOTE** *Long-term backups and Retention Lock, if required, must be requested via the infrastructure support process (<backup-support-group>), providing the ADB-S OCID and the required retention period.*

11.9 Manual Backup — What It Is and When to Use It

The manual backup in ADB-S Serverless is an on-demand backup that can be launched from the OCI Console at any time from the "Backups" tab → "Create manual backup", via API, or via OCI CLI. During the backup operation the database remains available — it does not go offline. Lifecycle management operations such as stop, scaling, and terminate are however disabled during backup. The typical use case: before a significant application or schema modification, a manual backup is launched as a safe restore point, before proceeding.

What is NOT possible with a manual backup: choose what to back up — it is always the entire database; choose where it goes — always Oracle-managed object storage; download it or access the storage directly; use it to migrate the database to another system via physical restore.

https://docs.oracle.com/en-us/iaas/api/#/en/database/20160918/AutonomousDatabaseBackup/CreateAutonomousDatabaseBackup
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/backup-restore-notes.html

11.10 IAM Policy for Backup and Restore

```
-- View database and initiate restore:
allow group ADB-Backup-Operators to use autonomous-databases
  in compartment <compartment-name>
-- Read backups (required for restore):
allow group ADB-Backup-Operators to read autonomous-backups
  in compartment <compartment-name>
-- Manage manual and long-term backups (if required):
allow group ADB-Backup-Operators to manage autonomous-backups
  in compartment <compartment-name>
-- Granular per-operation (least privilege):
allow group ADB-Backup-Operators to use autonomous-databases
  in compartment <compartment-name>
  where any {
    request.operation = 'RestoreAutonomousDatabase',
    request.operation = 'GetAutonomousDatabase',
    request.operation = 'ListAutonomousDatabases'
  }
allow group ADB-Backup-Operators to read autonomous-backups
  in compartment <compartment-name>
  where any {
    request.operation = 'GetAutonomousDatabaseBackup',
    request.operation = 'ListAutonomousDatabaseBackups'
  }
```

https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/autonomous-database-iam-policies.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/backup-restore.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/backup-intro.html

12. User Management, Profiles and IAM Database Token

12.1 IAM Database Token — Least Privilege Architecture

The implementation of ADB-S access via IAM Database Token allows completely centralizing identity and authentication in OCI IAM, eliminating the use of static credentials at the database level. The profile assigned to application SQL/JDBC users must implement the Least Privilege principle: IAM configuration must avoid using aggregate families and rely exclusively on atomic permissions assigned based on the real required operational scope.

IAM Database Token generation is a control plane operation invoked via oci iam db-token get. This phase requires authorization on the database-connections resource. Although database-connections is technically included in autonomous-database-family, a correct engineering approach requires explicitly declaring the specific permission, avoiding inheritance of unnecessary privileges:

```
-- Minimum for IAM Database Token generation:
allow group ADB-App-Developers to use database-connections in tenancy

-- Required for token handshake with ADB-S service:
allow group ADB-App-Developers to use autonomous-database in compartment <compartment-name>
WHERE ANY { request.operation = 'GetAutonomousDatabaseConsoleToken' }

-- NOTE: do NOT use autonomous-database-family — it includes administrative operations
-- inconsistent with an application profile.
```

IAM Database Token-based authentication replaces static database-level verification mechanisms with events completely tracked in OCI Audit, ensuring greater visibility and governance of accesses.

12.2 Interactive Tools — SQL Developer, SQLcl, SQL*Plus

ADB-S access via IAM Database Token is supported via Oracle interactive tools — SQL Developer (classic and with VS Code), SQLcl, SQL*Plus — using JDBC Thin driver and secure connections based on TCPS protocol, both in unilateral TLS (TCPS) and mutual TLS (mTLS) mode. The IAM Database Token represents the temporary credential used for logical authentication toward the Autonomous Database service, while the TLS protocol (TCPS or mTLS) is responsible exclusively for transport channel protection and authentication.

Oracle officially documents IAM Database Token usage with Autonomous Database in combination with command-line tools and JDBC clients including SQLcl and SQL*Plus. For SQL Developer, Oracle describes an evolved model based on next-generation JDBC driver, in which the client natively manages IAM Database Token acquisition and renewal, completely delegating JDBC layer interaction with OCI IAM.

In the specific context analyzed, the IAM Database Token auto-generation mechanism via SQL Developer is not currently supported in enterprise environments protected by HTTP/HTTPS proxy with NTLM authentication. Following verification with Oracle Support, this behavior was recognized as a product limitation and formalized via the opening of specific Enhancement Requests for both classic SQL Developer and SQL Developer for VS Code. In the presence of this constraint, IAM Database Token usage via interactive tools remains possible adopting a model where the token is generated externally and consumed by the JDBC client:

```
# Generate IAM Database Token via OCI CLI (required in NTLM proxy environments):
oci session authenticate          # opens browser, valid 1h, renewable 24h
oci iam db-token get --auth security_token

# Renew session:
oci session refresh --profile DEFAULT
```

12.3 M2M Scenarios — Machine-to-Machine

IAM Database Token authentication toward ADB-S is not limited to interactive tools but represents the reference mechanism for machine-to-machine (M2M) application scenarios. In all these contexts, the IAM Database Token constitutes a temporary credential issued by OCI IAM used exclusively for the authentication handshake toward the Autonomous Database service, while communication channel protection occurs via TCPS or mTLS. The transport mechanism (TCPS or mTLS) remains orthogonal to the token-based authentication model.

M2M Within OCI — Resource Principal / Instance Principal

In M2M scenarios executed within OCI, IAM Database Token acquisition typically occurs via Resource Principal or Instance Principal, completely eliminating the use of API Keys and static secrets. Applications on OCI Functions, OCI Container Instances, OCI Compute, or Oracle Kubernetes Engine (OKE) authenticate natively toward OCI IAM via the principal associated with the execution resource, invoke IAM APIs for Database Token generation, and use the token for ADB-S connection via JDBC driver (OJDBC Thin) or Oracle Python driver

(oracledb framework), on TCPS or mTLS channels. Access control is entirely delegated to IAM policies and principal scope, ensuring a security level consistent with Zero Trust requirements and the Least Privilege principle.

M2M External to OCI — On-Premise or Other Cloud

In M2M scenarios external to OCI (e.g. on-premise applications or workloads on other cloud providers), IAM Database Token acquisition requires authentication toward OCI IAM via API Key, configured with Least Privilege policies. The API Key is never used to access the database directly, but exclusively to authenticate toward OCI IAM and request a temporary token issuance. The IAM Database Token is then consumed by the application via OJDBC (Java) or Oracle Python driver for ADB-S connection, always on TCPS or mTLS channels, maintaining clear separation between the identity plane and the database access plane.

12.4 Security Architecture Summary

In all described models — interactive tools, M2M in OCI, external M2M — the IAM Database Token and associated IAM policies represent the unified authentication mechanism toward Autonomous Database Serverless, while API Key, Resource Principal, Instance Principal, or CLI tools constitute exclusively mechanisms for authentication toward OCI IAM, necessary for token issuance and not alternative to it. This approach standardizes ADB-S access, improves access traceability via OCI Audit, and significantly reduces the exposure surface associated with static application credentials, independently of the TCPS or mTLS connection mode adopted.

The adoption of IAM Database Token introduces a security model in which application authentication, authorization, and database access are clearly separated and governed via OCI IAM. The token is a temporary credential with limited validity, issued on the OCI IAM control plane and used exclusively for the authentication handshake toward the Autonomous Database service plane, on TCPS or mTLS connections. Applications do not retain persistent passwords or database authentication certificates, but use exclusively temporary tokens whose lifecycle is governed by OCI IAM and traced via OCI Audit. The architecture enforces separation between: application authentication toward OCI IAM; temporary token issuance; encrypted database access via TCPS or mTLS; complete operation traceability via OCI Audit. This approach satisfies enterprise security requirements and compliance requirements, maintaining a standardized and consistent access model for both interactive tools and machine-to-machine application scenarios.

13. On Boarding Process

The On Boarding process describes the steps necessary to allow an application team (<application-owner-group>) requesting the ADB-S service to be able to use it. The operations that will be performed are the following:

- Step 1 — Request: The applicant (<application-owner-group>) requests the service from the Design team representative of <cloud-provider> providing among other things the name of the key-secret resource (already created in the Compartment where ADB-S will reside) to associate with the ADB-S instance for TDE encryption.
- Step 2 — Design: The Design team inserts in the LLD document the characteristics of resources to be created on OCI, lists the application profiles to be created on <organization> IAM to be associated with the application initiative, mapped onto the relevant authorization profiles predisposed on OCI IAM.
- Step 3 — Delivery: The Delivery group of <cloud-provider> proceeds with provisioning OCI resources (associating the new ADB-S with the key to create the TDE already defined on OCI by the <application-owner-group>) and in parallel the Cloud Architectures team engages the <iam-ops-team> for creating profiles on <organization> IAM of application user groups, and creates the relevant user groups on OCI IAM.
- Step 4 — Asset Registration: The ADB-S resource is entered in Asset. Assetization as a logical object with the following attributes: [compartment-name].[adb-s-name] and PE private IP (add network/netmask in the notes field).
- Step 5 — Delivery Release: Delivery is released to the requestor.

14. Pricing

14.1 Backup Management — Three Typologies

Backup management in Oracle Autonomous Database Serverless is a fundamental pillar for business continuity and data protection, articulated in three main typologies operating on different persistence and flexibility levels.

Automatic Backups represent the first line of defense: they are executed daily by Oracle without any user intervention, guaranteeing a minimum Recovery Point Objective (RPO). These backups are incremental and are retained in Oracle-managed Object Storage for a period (retention) that the user can configure between 1 and 60 days. If the database is configured with Autonomous Data Guard, automatic backups are also replicated in the standby region to maximize resilience. Oracle includes a standard backup quota free of charge (typically equal to allocated storage in the OCPU model, or a base window in the ECPU model), while the excess is billed based on GB-month consumed.

Manual Backups (On-Demand) intervene when it is necessary to create an immediate restore point, for example before a critical applicative operation or structural schema modification. These backups can be started from the OCI console, via API, or via OCI CLI, and are stored in the same Object Storage as automatic backups. Their retention duration is tied to the retention policy set for automatic backups: if retention is 30 days, the manual backup will also expire after that period. Manual backups cannot be individually deleted by the user but follow the automatic lifecycle of the system.

Long-Term Backups (LTB) satisfy requirements of historical archiving or legal compliance (GDPR, banking regulations). Unlike the previous ones, LTBs are independent "Full" backups that allow retaining the database state from 3 months to 10 years. The user can create a Long-Term Backup manually or configure automatic scheduling with weekly, monthly, or yearly frequency. These backups are not included in the free quota and are always billed as additional storage. If an Autonomous Database instance is deleted, the user has the possibility to preserve the related Long-Term Backups, allowing database restoration at a future time on a new instance.

https://docs.oracle.com/en-us/iaas/autonomous-database-shared/doc/autonomous-backup.html
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/backup-long-term.html
https://www.oracle.com/cloud/price-list/

14.2 Backup Billing Summary

Autonomous Database Serverless implements a data protection system based on backups automatically managed by Oracle. Backups are immutable, cannot be modified or manually deleted, and are automatically replicated between Availability Domains or Fault Domains depending on region configuration, guaranteeing high data durability and operational continuity.

Automatic backups execute periodic backups of type full, cumulative, and incremental and maintain a continuous transaction backup system enabling a Recovery Point Objective (RPO) of less than 10 seconds and Point-in-Time Recovery up to a maximum of 60 days. In the OCPU model, automatic backup storage is included in the database storage cost; in the ECPU model, backup storage is billed separately as backup storage beyond the database storage.

In the OCI cost report, backup storage appears as storage associated with the Autonomous Database service. The console shows the aggregate value in the "Total Backup Storage" field, which includes automatic backup storage (ECPU model) and Long-Term Backup storage.

14.3 ADB-S Pricing Metrics — ECPU Model

Autonomous Database Serverless uses a pay-as-you-go billing model based on the ECPU (Elastic CPU) metric, introduced by Oracle to provide more granular and real-consumption-adherent measurement compared to the previous OCPU model. The service cost is determined exclusively by actual resource usage, without fixed pre-allocated capacities.

Compute (ECPU): The compute component is billed in ECPU-hours, calculated as the product of the number of ECPUs actually used and usage time. ECPUs represent elastic compute capacity dynamically allocated based on workload. With auto-scaling enabled, the database can automatically increase ECPUs beyond the base value up to the configured maximum limit. Additional resources are billed only for the period in which they are actually used, with sub-hourly granularity based on effective average consumption. No fixed costs or over-provisioning. The unit price of ECPU is all-inclusive and includes: software license (License Included); automatic database management (patching, tuning, high availability); native security features including Transparent Data Encryption and basic auditing.

Storage and Backup: In the ECPU model, storage is separate from compute, enabling more accurate cost control. Database storage is billed separately in GB/month for ATP/JSON/APEX workloads and in TB/month for ADW.

Storage is elastic and can grow automatically. Automatic backup storage is a distinct cost item billed in GB with configurable retention, allowing paying exclusively for space actually occupied by retained backups.

14.4 Data Safe Pricing Metrics

Oracle Data Safe (Audit Trail Collection) uses a pay-as-you-go billing model based on the volume of audit records collected from monitored target databases. The billing metric is Audit Records Per Target Database Per Month. For each database registered in Data Safe a free threshold of 1,000,000 audit records per month per target database is included. Only records exceeding this threshold generate cost.

Beyond the free threshold, additional records are billed per the unit of measure 10,000 Audit Records Per Target Per Month at unit price €0.093. The cost is calculated exclusively on records beyond the first monthly million per target database, corresponding to approximately €9.30 per additional million audit records. This model allows maintaining zero cost for databases with low auditing volume, while databases with extended audit policies or high-activity workloads can generate volumes exceeding the free threshold.

In the OCI Cost and Usage Report, consumption appears as Data Safe – Audit Records Collection service usage with hourly time granularity. The report shows the volume of records ingested and directly expose the target database generating the consumption. In the OCI Cost and Usage Report, consumption appears with **product/service** = Data Safe and hourly time granularity. Each registered target database appears as a distinct row identified by its **product/resourceId**, allowing direct cost attribution per target..

https://www.oracle.com/cloud/price-list/
https://docs.oracle.com/en/cloud/paas/data-safe/udscs/audit-profiles.html

15. Support

To request support for issues encountered during use or implementation of the service, the following features are available in the OCI Console:

- Create a Support Request — for service-specific issues requiring Oracle Cloud Infrastructure support
- Access to the Support Center — for case management and Oracle Cloud Status

NOTE *For non-applicative but infrastructure-level issues, contact the Competence Center by opening an incident ticket via the IT service management platform toward the infrastructure support group (<infra-support-group>).*

Appendix — Official Reference Links

ADB-S Documentation

https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/index.html
https://docs.oracle.com/en-us/iaas/Content/Audit/Concepts/auditoverview.htm
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/adb-audit.html
https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/AUDIT-Unified-Auditing.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/adb-audit.html
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-monitor-metrics-list.html
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-database-events.html
https://docs.oracle.com/en-us/iaas/autonomous-database-shared/doc/autonomous-backup.html
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/backup-long-term.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/backup-restore.html
https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/autonomous-database-iam-policies.html
https://docs.oracle.com/en-us/iaas/api/#/en/database/20160918/AutonomousDatabaseBackup/CreateAutonomousDatabaseBackup
https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/pipeline-oracle-maintained.html

Data Safe Documentation

https://docs.oracle.com/en/cloud/paas/data-safe/udscs/audit-trails.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/permissions-register-autonomous-database.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/create-iam-policies-oracle-data-safe-users.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/data-safe-family-resource.html
https://docs.oracle.com/en-us/iaas/data-safe/doc/roles-oracle-data-safe-service-account.html
https://docs.oracle.com/en-us/iaas/data-safe/doc/view-and-manage-audit-trails.html
https://docs.oracle.com/en/cloud/paas/data-safe/udscs/alerts-overview.html
https://docs.oracle.com/en/cloud/paas/data-safe/admds/event-types-oracle-data-safe.html
https://docs.oracle.com/en/learn/datasafe-audit-log-to-oci-logging/index.html
https://docs.oracle.com/en-us/iaas/data-safe/doc/target-database-registration-overview.html
https://docs.oracle.com/en/cloud/paas/data-safe/udscs/using-oracle-data-safe.pdf
https://blogs.oracle.com/database/data-safe-notifications

SYS_CONTEXT and Pricing

https://docs.oracle.com/en/database/oracle/oracle-database/26/sqlrf/SYS_CONTEXT.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/dbseg/creating-custom-unified-audit-policies.html
https://docs.oracle.com/en/cloud/paas/data-safe/udscs/audit-profiles.html
https://www.oracle.com/cloud/price-list/